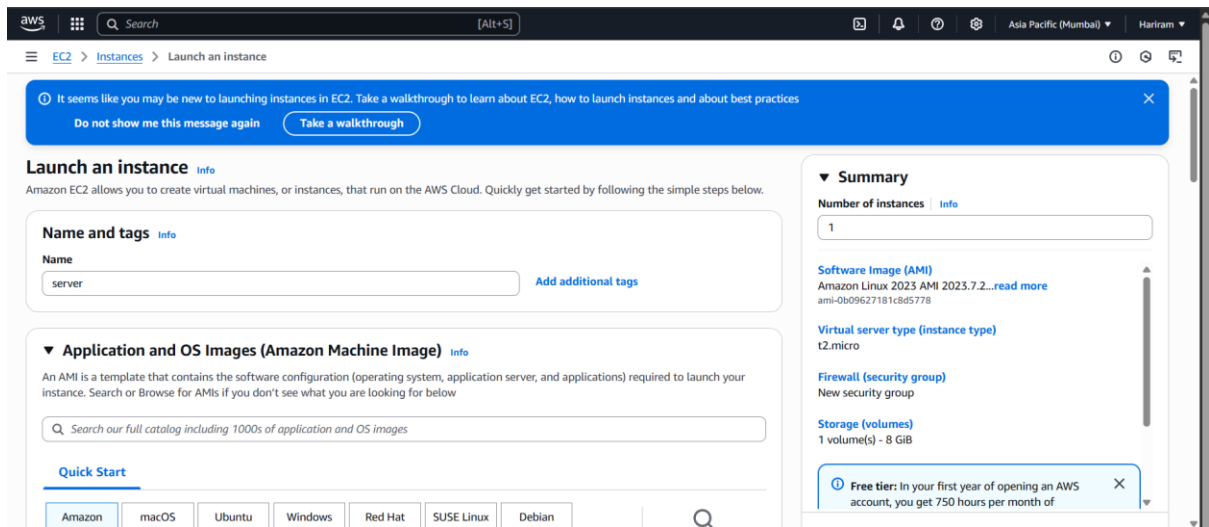
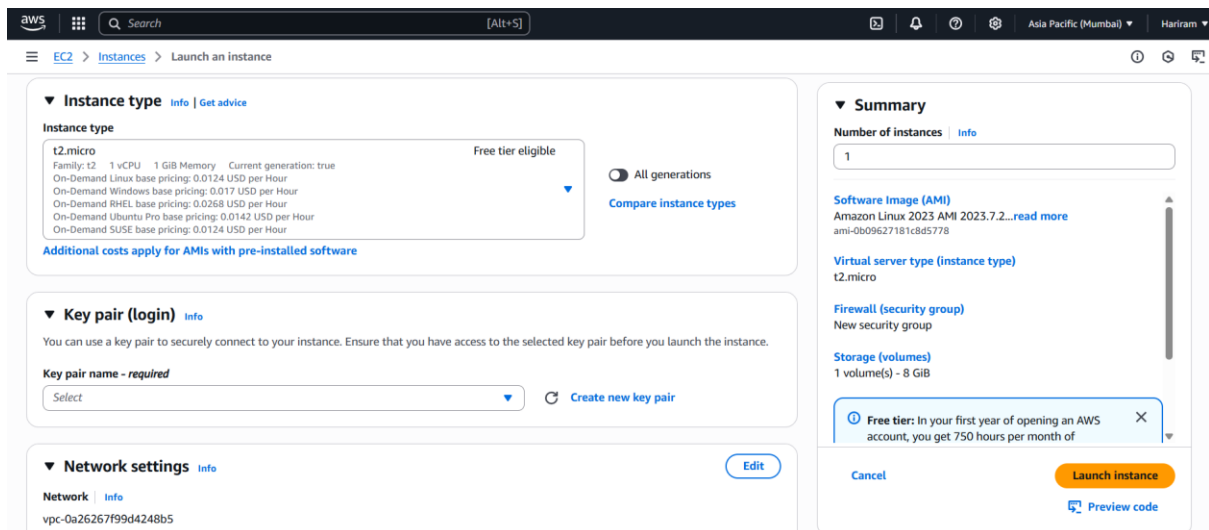


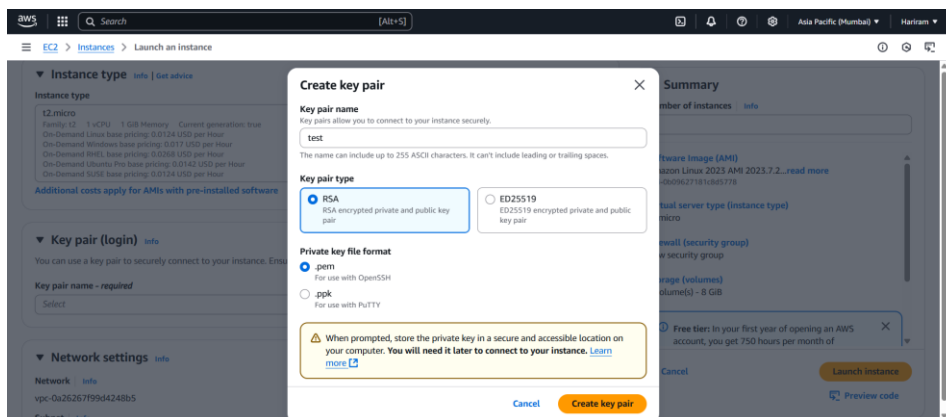
- 1) Deploy a Ec2 instance using of t2. micro instance in Mumbai Region in Zone ap-south-1a, Configure web server on it and make it live. and attach a 5gb ebs volume. where create 10 files. like training.txt



EC2 instance is created with name server.



New key is created .



Network settings

VPC - required [Info](#)
vpc-0a26267f99d4248b5 (default)

Subnet - required [Info](#)
subnet-0c0cb1aef6691d743
VPC: vpc-0a26267f99d4248b5 | Owner: 163083073206 | Availability Zone: ap-south-1a | IP addresses available: 4091 | CIDR: 172.31.32.0/20

Auto-assign public IP - required [Info](#)
Enable

Additional charges apply when outside of free tier allowance

Firewall (security groups) - required [Info](#)
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

[Create security group](#) [Select existing security group](#)

Security group name - required
dev-sg
This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and _ (underscore) and hyphen (-).

Description - required [Info](#)
devsg created 2025-06-17T08:33:53.631Z

Inbound Security Group Rules

Security group rule 1 (TCP: 22, 0.0.0.0/0)

Type [Info](#) | Protocol [Info](#) | Port range [Info](#)

ssh | TCP | 22

Source type [Info](#) | Source [Info](#) | Description - optional [Info](#)

Anywhere | 0.0.0.0/0 | e.g. SSH for admin desktop

[Remove](#)

Summary

Number of instances [Info](#)
1

Software Image (AMI)
Amazon Linux 2023.7.2...[read more](#)
ami-0b09627181c8d5778

Virtual server type (instance type)
t2.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet.

New security group is created.

devsg created 2025-06-17T08:33:53.631Z

Inbound Security Group Rules

Security group rule 1 (TCP: 22, 0.0.0.0/0)

Type [Info](#) | Protocol [Info](#) | Port range [Info](#)

ssh | TCP | 22

Source type [Info](#) | Source [Info](#) | Description - optional [Info](#)

Anywhere | 0.0.0.0/0 | e.g. SSH for admin desktop

[Remove](#)

Security group rule 2 (TCP: 80, 0.0.0.0/0)

Type [Info](#) | Protocol [Info](#) | Port range [Info](#)

HTTP | TCP | 80

Source type [Info](#) | Source [Info](#) | Description - optional [Info](#)

Anywhere | 0.0.0.0/0 | e.g. SSH for admin desktop

[Remove](#)

Security group rule 3 (ICMP: All, 0.0.0.0/0)

Type [Info](#) | Protocol [Info](#) | Port range [Info](#)

All ICMP - IPv4 | ICMP | All

Source type [Info](#) | Source [Info](#) | Description - optional [Info](#)

Anywhere | 0.0.0.0/0 | e.g. SSH for admin desktop

[Remove](#)

Summary

Number of instances [Info](#)
1

Software Image (AMI)
Amazon Linux 2023.7.2...[read more](#)
ami-0b09627181c8d5778

Virtual server type (instance type)
t2.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet.

Necessary networks are allowed.

Success
Successfully initiated launch of instance (i-049da72534d999cec)

[Launch log](#)

Instances is created successfully.

EC2 [Dashboard](#) [EC2 Global View](#) [Events](#) **Instances**

Instances (1/1) [Info](#)

Last updated less than a minute ago [Refresh](#) [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Find Instance by attribute or tag (case-sensitive) [All states](#)

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IP
☒	server	i-049da72534d999cec	Running	t2.micro	Initializing	View alarms	ap-south-1a	ec2-13-2f...

Instances running successfully.

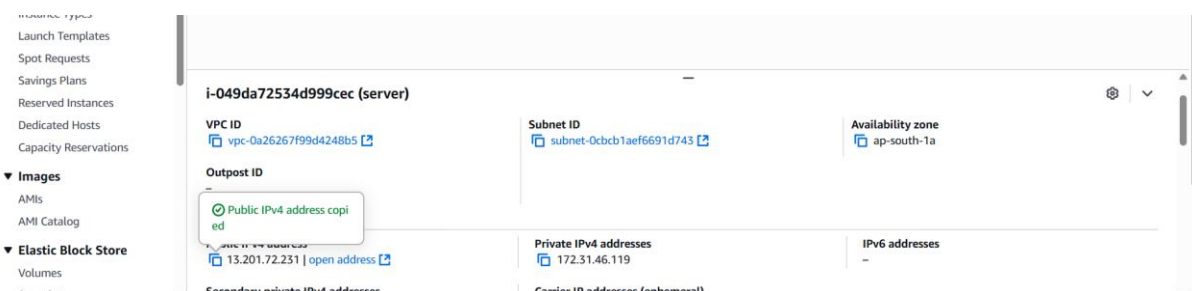
[illegible]

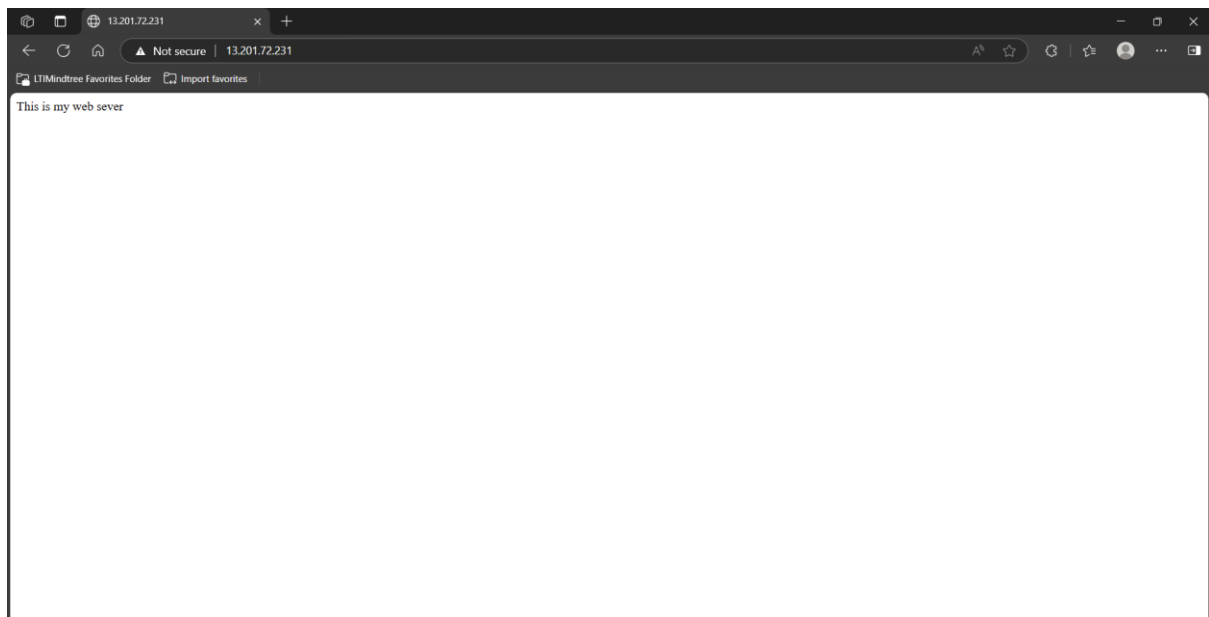
```
[ec2-user@ip-172-31-46-119 ~]$ sudo su -
[root@ip-172-31-46-119 ~]# hostnamectl set-hostname server.example.com
[root@ip-172-31-46-119 ~]# bash
[root@server ~]#
[root@server ~]# yum install httpd -y
Amazon Linux 2023 Kernel Livepatch repository           156 kB/s | 17 kB      00:00
Dependencies resolved.

=====
Package                                Architecture      Version                                Repository      Size
=====
Installing:
httpd                                   x86_64            2.4.62-1.amzn2023                        amazonlinux      48 k
Installing dependencies:
apr                                     x86_64            1.7.5-1.amzn2023.0.4                    amazonlinux      129 k
apr-util                               x86_64            1.6.3-1.amzn2023.0.1                    amazonlinux      98 k
generic-logos-httpd                   noarch            18.0.0-12.amzn2023.0.3                  amazonlinux      19 k
httpd-core                             x86_64            2.4.62-1.amzn2023                        amazonlinux      1.4 M
httpd-filesystem                       noarch            2.4.62-1.amzn2023                        amazonlinux      14 k
httpd-tools                            x86_64            2.4.62-1.amzn2023                        amazonlinux      81 k
libbrotli                               x86_64            1.0.9-4.amzn2023.0.2                    amazonlinux      315 k
mailcap                                noarch            2.1.49-3.amzn2023.0.3                    amazonlinux      33 k
=====
```

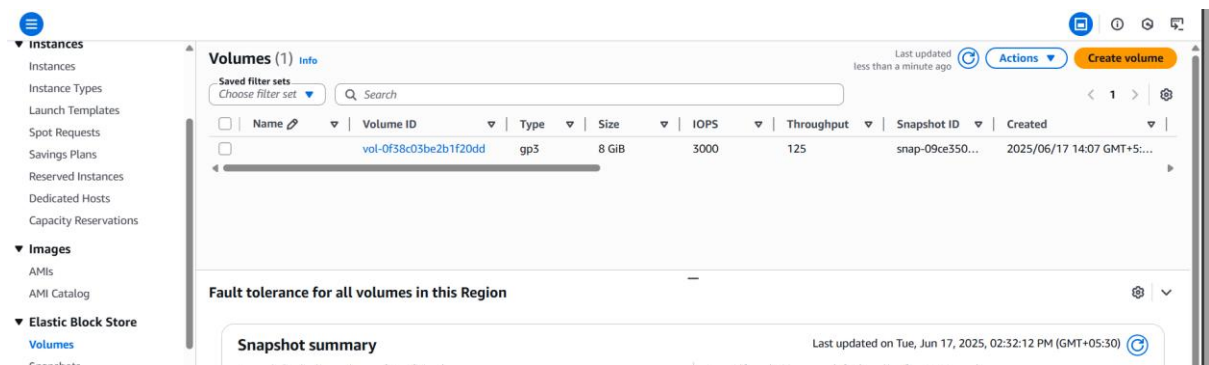
```
Complete!
[root@server ~]# cd /var/www/html
[root@server html]# echo "This is my web sever" > index.html
[root@server html]# ll
total 4
-rw-r--r--. 1 root root 21 Jun 17 08:58 index.html
[root@server html]# cat index.html
This is my web sever
[root@server html]# systemctl restart httpd
[root@server html]# systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[root@server html]# |
```

Created a html file to host in server.





Web server is live now .



New EBS volume is created.

Create volume [Info](#)
Create an Amazon EBS volume to attach to any EC2 instance in the same Availability Zone.

Volume settings

Volume type [Info](#)
General Purpose SSD (gp3)

Size (GiB) [Info](#)
5
Min: 1 GiB, Max: 16384 GiB.

IOPS [Info](#)
3000
Min: 3000 IOPS, Max: 16000 IOPS.

Throughput (MiB/s) [Info](#)
125
Min: 125 MiB, Max: 1000 MiB, Baseline: 125 MiB/s.

Availability Zone [Info](#)
ap-south-1a

Snapshot ID - optional [Info](#)
Don't create volume from a snapshot

Encryption [Info](#)
☐ Encrypt this volume
Use Amazon EBS encryption as an encryption solution for your EBS resources associated with your EC2 instances.

Tags - optional [Info](#)
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.
No tags associated with the resource.
[Add tag](#)
You can add 50 more tags.

Snapshot summary [Info](#) [Refresh](#)
[Click refresh to view backup information](#)
The volume type that you select and the tags that you assign determine whether the volume will be backed up by any Data Lifecycle Manager policies.

[Cancel](#) [Create volume](#)

Volume setting is updated as 5gb .

Volumes (2) [Info](#)

Last updated less than a minute ago [Actions](#) [Create volume](#)

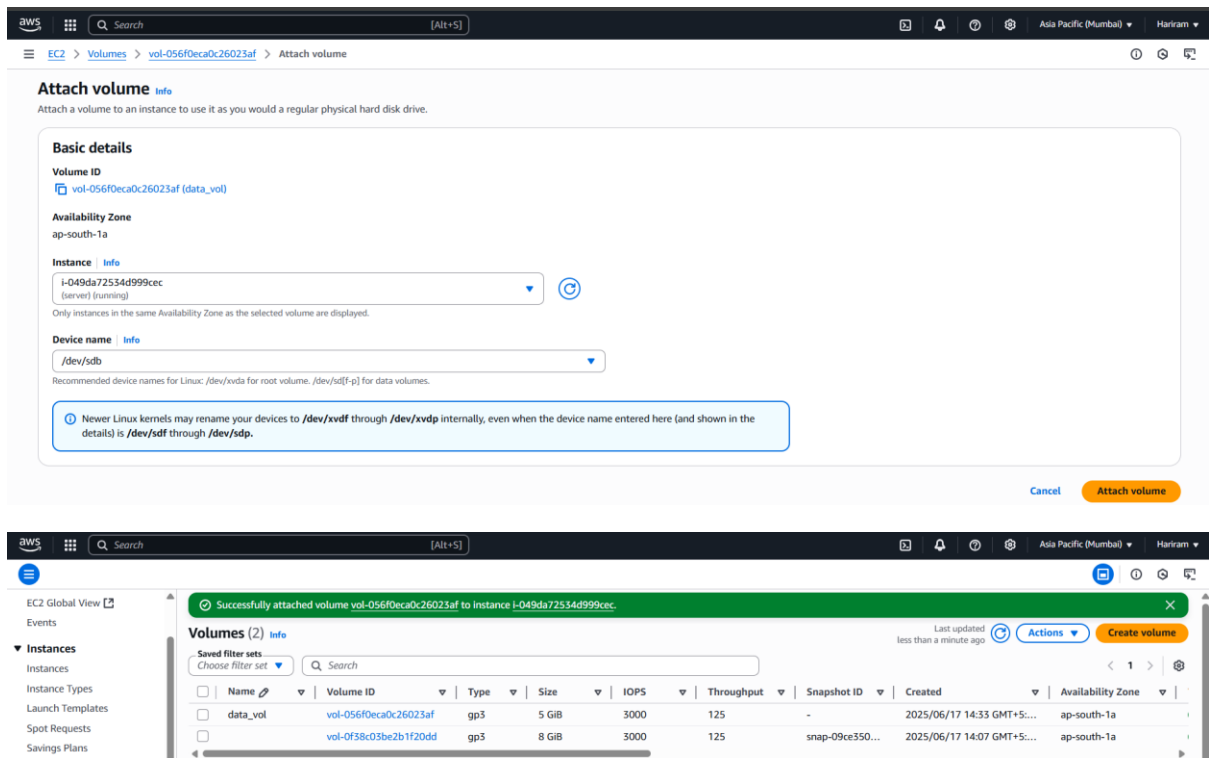
☐	Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot ID	Created	Availability Zone
☐		vol-056f0eca0c26023af	gp3	5 GiB	3000	125	-	2025/06/17 14:33 GMT+5:...	ap-south-1a
☐		vol-0f38c03be2b1f20dd	gp3	8 GiB	3000	125	snap-09ce350...	2025/06/17 14:07 GMT+5:...	ap-south-1a

Volume is created successfully.

Volumes (1/2) [Info](#)

Last updated less than a minute ago [Actions](#) [Create volume](#)

☒	Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot ID	Created	Availability Zone
☒	data_vol	vol-056f0eca0c26023af	gp3	5 GiB	3000	125	-	2025/06/17 14:33 GMT+5:...	ap-south-1a



Volume is connected to the Instance successfully.

```
[root@server ~]# lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
xvda        202:0    0   8G  0 disk
├─xvda1     202:1    0   8G  0 part /
├─xvda127   259:0    0    1M  0 part
├─xvda128   259:1    0   10M  0 part /boot/efi
└─xvdb      202:16   0    5G  0 disk
[root@server ~]# mkfs.ext4 /dev/sdb
mke2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 1310720 4k blocks and 327680 inodes
Filesystem UUID: 01129d00-ab73-47b7-bb79-a0a17d3c47c1
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done
```

Volume is formatted and file system is created.

```
[root@server ~]# mkdir /data
[root@server ~]# mount /dev/xvdb /data/
[root@server ~]# df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        4.0M    0  4.0M   0% /dev
tmpfs           475M    0  475M   0% /dev/shm
tmpfs           190M  456K  190M   1% /run
/dev/xvda1       8.0G  1.6G   6.4G  20% /
tmpfs           475M    0  475M   0% /tmp
/dev/xvda128     10M  1.3M   8.7M  13% /boot/efi
tmpfs           95M    0   95M   0% /run/user/1000
/dev/xvdb        4.9G  24K   4.6G   1% /data
[root@server ~]# cd /data
[root@server data]#
```

Data directory is created and mounted with the volume.

```
[root@server data]# touch training.txt{1..10}
[root@server data]# ll
total 16
drwx-----. 2 root root 16384 Jun 17 09:05 lost+found
-rw-r--r--. 1 root root      0 Jun 17 09:07 training.txt1
-rw-r--r--. 1 root root      0 Jun 17 09:07 training.txt10
-rw-r--r--. 1 root root      0 Jun 17 09:07 training.txt2
-rw-r--r--. 1 root root      0 Jun 17 09:07 training.txt3
-rw-r--r--. 1 root root      0 Jun 17 09:07 training.txt4
-rw-r--r--. 1 root root      0 Jun 17 09:07 training.txt5
-rw-r--r--. 1 root root      0 Jun 17 09:07 training.txt6
-rw-r--r--. 1 root root      0 Jun 17 09:07 training.txt7
-rw-r--r--. 1 root root      0 Jun 17 09:07 training.txt8
-rw-r--r--. 1 root root      0 Jun 17 09:07 training.txt9
[root@server data]#
```

Text file is created inside the data volume.

2) We people are working on a common project in a same region. But my servers are in different zones. So, I want to share project information with everyone simultaneously. Configure efs storage it should be mount on every server.

Create file system

Create a file system with the recommended settings shown below by choosing Create file system. To view all settings or to customize your file system, choose Customize. [Learn more](#)

Name - optional
Name your file system.

Name can include letters, numbers, and +, -, _, / symbols, up to 256 characters.

Virtual Private Cloud (VPC)
Choose the VPC where you want EC2 instances to connect to your file system.

Recommended settings
Your file system is created with the following recommended settings unless you choose to customize the file system. You will be charged for storage and throughput. We recommend reviewing pricing for these features using the [AWS Pricing Calculator](#).

Setting	Value	Editable after creation
Throughput mode Learn more	Elastic	Yes
Transition into Infrequent Access (IA)	30 day(s) since last access	Yes

Cancel Customize **Create file system**

EFS file system is created.

Success!
File system (fs-0ee5cf7076c3840ba) is available. [View file system](#)

File systems (1) [Refresh](#) [View details](#) [Delete](#) [Create file system](#)

Name	File system ID	Encrypted	Total size	Size in Standard	Size in IA	Size in Archive	Provisioned Throughput (MiB/s)
MY-EFS	fs-0ee5cf7076c3840ba	Encrypted	6.00 KiB	6.00 KiB	0 Bytes	0 Bytes	-

Successfully EFS is created.

Network [Refresh](#) [Manage](#)

Availability zone (AZ-ID)	Mount target ID	Subnet ID	Mount target state	IPv4 address	IPv6 address	Network interface ID	Security groups
us-east-1a (use1-az4)	fsmt-04ebffa6f6be296819	subnet-00608da34c56e569d	Creating	172.31.27.235	N/A	eni-05280bbf7b35c05f9	-
us-east-1b (use1-az6)	fsmt-0aa41db09736171d2	subnet-0541b7ff15bd92fd9	Creating	172.31.32.238	N/A	eni-0cd5381d5c6bc53a4	-
us-east-1c (use1-az1)	fsmt-0f7afe7bd443f4362	subnet-059fb42128e9ff90	Creating	172.31.11.123	N/A	eni-0a975832ed32a50bb	-
us-east-1d (use1-az2)	fsmt-0ee8f004f7358fd5b	subnet-0d4f0214153564e39	Creating	172.31.81.136	N/A	eni-0118e1635d313f3ca	-

Waiting for EFS to create file system in every zone.

Create security group [Info](#)

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name [Info](#)
my-nfs
Name cannot be edited after creation.

Description [Info](#)
allow nfs

VPC [Info](#)
vpc-0c1e9f84aa49d9d9e

Inbound rules [Info](#)

This security group has no inbound rules.

[Add rule](#)

New security group is created as my-nfs.

Outbound rules [Info](#)

Type [Info](#) Protocol [Info](#) Port range [Info](#) Destination [Info](#) Description - optional [Info](#)

NFS TCP 2049 Custom 0.0.0.0/0 [Delete](#)

[Add rule](#)

Rules with destination of 0.0.0.0/0 or ::/0 allow your instances to send traffic to any IPv4 or IPv6 address. We recommend setting security group rules to be more restrictive and to only allow traffic to specific

Outbound rules is updated with NFS.

Security group (sg-026903ce67b7a389d | my-nfs) was created successfully

sg-026903ce67b7a389d - my-nfs [Actions](#)

Details

Security group name my-nfs	Security group ID sg-026903ce67b7a389d	Description allow nfs	VPC ID vpc-0c1e9f84aa49d9d9e
Owner 163083073206	Inbound rules count 0 Permission entries	Outbound rules count 1 Permission entry	

Security group created successfully.

Elastic File System

Amazon EFS > **File systems** > **fs-0ee5cf7076c3840ba** > **Network access**

Network

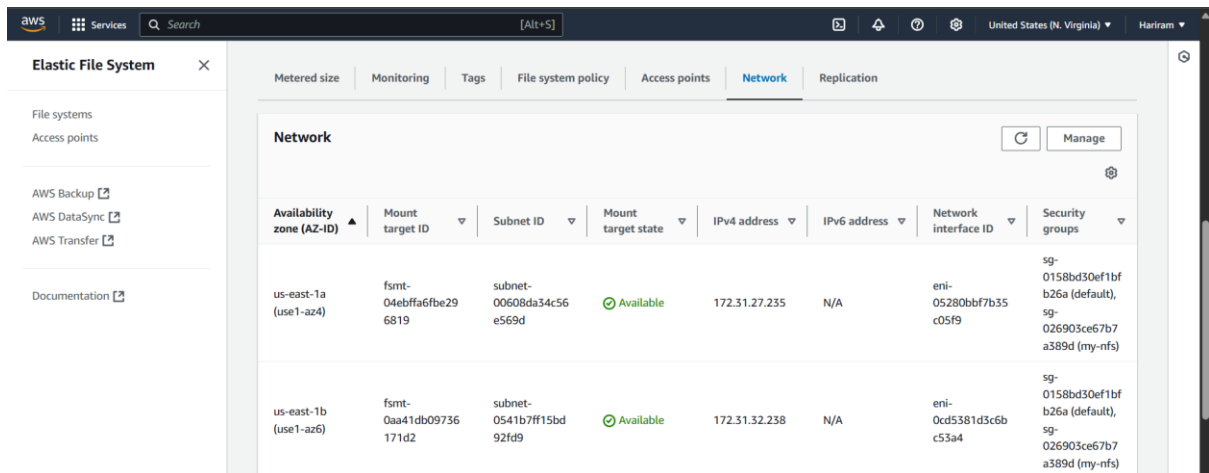
Virtual Private Cloud (VPC)
Choose the VPC where you want EC2 instances to connect to your file system.
vpc-0c1e9f84aa49d9d9e
You must delete all existing mount targets in order to change the VPC of your file system.

Mount targets
A mount target provides an NFSv4 endpoint at which you can mount an Amazon EFS file system. We recommend creating one

Availability zone	Subnet ID	IP address type	IPv4 address	IPv6 address
us-east-1a	subnet-00608d...	IPv4 only	172.31.27.235	-
us-east-1b	subnet-0541b7...	IPv4 only	172.31.32.238	-

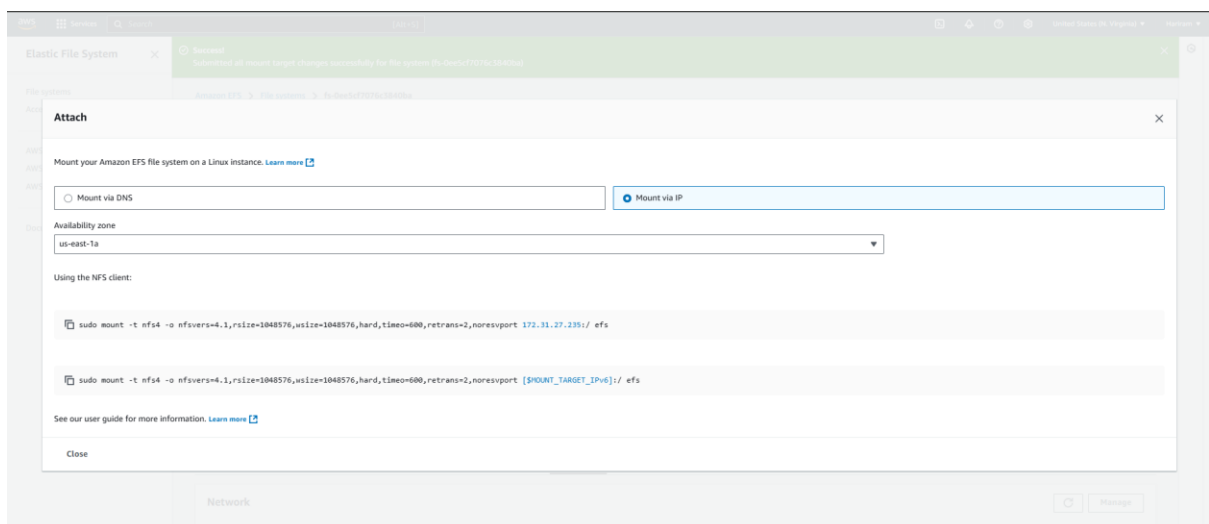
Dropdown menu options:
☐ sg-03cb5aa168bb5d450 launch-wizard-2
☒ sg-0158bd30ef1bf26a default
☐ sg-0175739b352878c9f efs-sg
☐ sg-0cd9a2b14bb27dd1a launch-wizard-1
☒ sg-026903ce67b7a389d my-nfs

Security group is add to every zone .



Availability zone (AZ-ID)	Mount target ID	Subnet ID	Mount target state	IPv4 address	IPv6 address	Network interface ID	Security groups
us-east-1a (use1-az4)	fsmt-04ebffa6f8e296819	subnet-00608da34c56e569d	Available	172.31.27.235	N/A	eni-05280bbf7b35c05f9	sg-0158bd30ef1bf b26a (default), sg-026903ce67b7 a389d (my-nfs)
us-east-1b (use1-az6)	fsmt-0aa41db09736171d2	subnet-0541b7ff15bd92fd9	Available	172.31.32.238	N/A	eni-0cd5381d3c6bc53a4	sg-0158bd30ef1bf b26a (default), sg-026903ce67b7 a389d (my-nfs)

EFS is successfully created to share project information with everyone simultaneously.



Mount your Amazon EFS file system on a Linux instance. [Learn more](#)

☐ Mount via DNS ☒ Mount via IP

Availability zone: us-east-1a

Using the NFS client:

```
sudo mount -t nfs4 -o nfsvers=4.1,rsize=1048576,wsz=1048576,hard,timeo=600,retrans=2,noresvport 172.31.27.235:/ efs
```

See our user guide for more information. [Learn more](#)

Close

Using this EFS can be attached.

Amazon and Redhat instance is created .

3) Create a Custom VPC. Where you need to create two subnets like Private subnet and Public subnet. In the public subnet, I want to host my web server. Where my website is running and Private subnet my database is running. Data base should not be reachable publicly.

The screenshot shows the 'Create VPC' page in the AWS Management Console. The page title is 'Create VPC' with an 'Info' link. Below the title, it states: 'A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.' The 'VPC settings' section includes 'Resources to create' with 'VPC only' selected, a 'Name tag - optional' field with 'my-vpc', and 'IPv4 CIDR block' set to '10.0.0.0/16'. The 'IPv6 CIDR block' section has 'No IPv6 CIDR block' selected. The 'Tenancy' is set to 'Default'. The 'Tags' section shows a key 'Name' and value 'my-vpc'. At the bottom, there are 'Cancel', 'Preview code', and 'Create VPC' buttons.

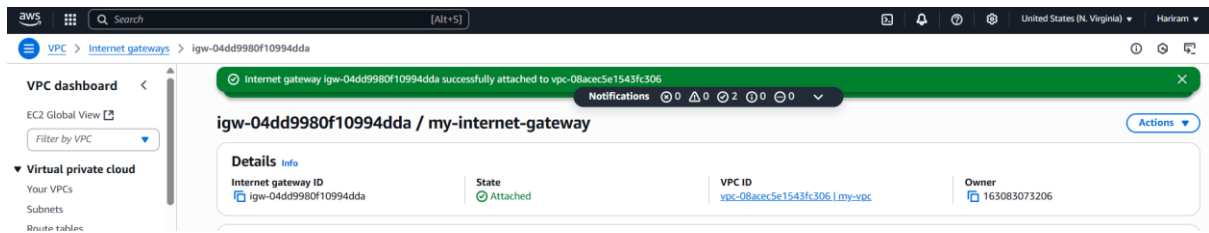
VPC is created with necessary settings. CIDR as 10.0.0.0.16

The screenshot shows the 'Create internet gateway' page in the AWS Management Console. The page title is 'Create internet gateway' with an 'Info' link. Below the title, it states: 'An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.' The 'Internet gateway settings' section includes a 'Name tag' field with 'my-internet-gateway'. The 'Tags - optional' section shows a key 'Name' and value 'my-internet-gateway'. At the bottom, there are 'Cancel' and 'Create internet gateway' buttons.

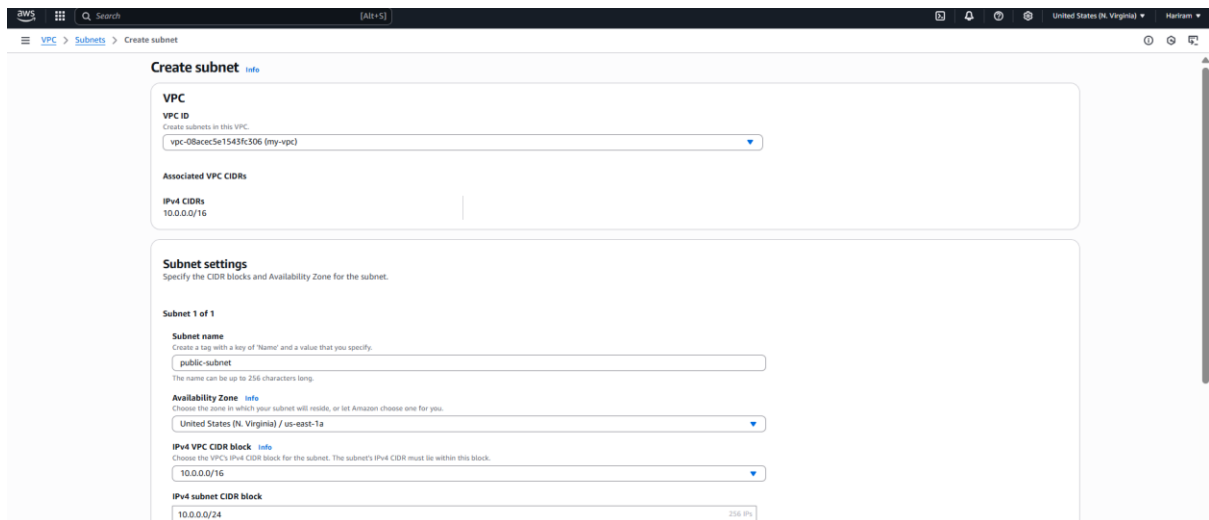
Internet gateway is created .

The screenshot shows the 'Attach to VPC' page in the AWS Management Console. At the top, a green notification bar states: 'The following internet gateway was created: igw-04dd9980f10994dda - my-internet-gateway. You can now attach to a VPC to enable the VPC to communicate with the internet.' The page title is 'Attach to VPC (igw-04dd9980f10994dda)' with an 'Info' link. Below the title, it states: 'Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.' The 'Available VPCs' section shows a search bar with 'vpc-08ace5e1543fc306'. At the bottom, there are 'Cancel' and 'Attach internet gateway' buttons.

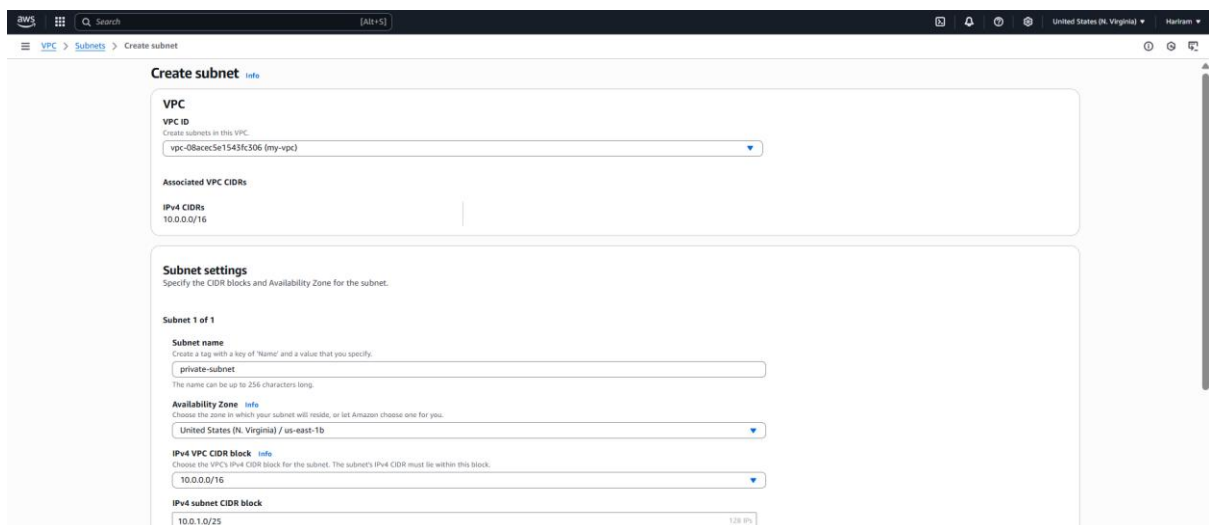
Internet gateway is attached with the VPC.



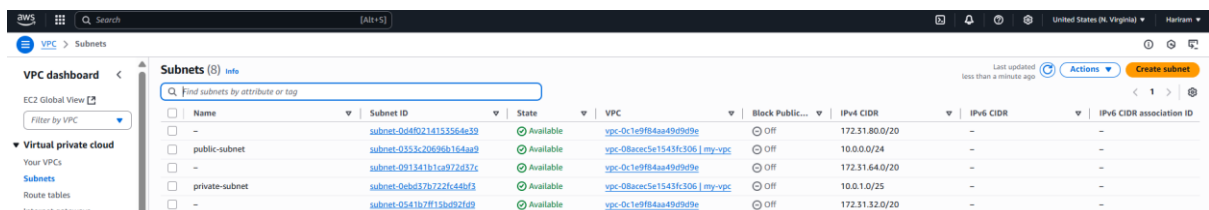
Successfully attached.



Public subnet is created with necessary settings. For public subnet CIDR : 10.0.0.0/24



Private subnet is created with necessary settings. For private subnet CIDR : 10.0.1.0/25



Public and private subnet are created successfully.

🔍 Search [Alt+S] United States (N. Virginia) 📍

VPC > NAT gateways > Create NAT gateway

🟢 Elastic IP address 44.207.228.221 (eipalloc-096223593032ecbba) allocated.

Create NAT gateway [Info](#)

A highly available, managed Network Address Translation (NAT) service that instances in private subnets can use to connect to services in other VPCs, on-premises networks, or the Internet.

NAT gateway settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

my-nat-gateway

The name can be up to 255 characters long.

Subnet
Select a subnet in which to create the NAT gateway.

subnet-0353c20696b164aa9 (public-subnet)

Connectivity type
Select a connectivity type for the NAT gateway.

☒ Public
☐ Private

Elastic IP allocation ID [Info](#)
Assign an Elastic IP address to the NAT gateway.

eipalloc-096223593032ecbba [Allocate Elastic IP](#)

▶ **Additional settings** [Info](#)

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Q Name X

Value - optional

Q my-nat-gateway X Remove

[Add new tag](#)

You can add 49 more tags.

[Cancel](#) [Create NAT gateway](#)

NAT gateway is created .

🔍 Search [Alt+S] United States (N. Virginia) 📍

VPC > Route tables > Create route table

Create route table [Info](#)

A route table specifies how packets are forwarded between the subnets within the VPC, the Internet, and your VPN connection.

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

public-route

VPC
The VPC to use for this route table.

vpc-08ace5e1543fc306 (my-vpc)

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Q Name X

Value - optional

Q public-route X Remove

[Add new tag](#)

You can add 49 more tags.

[Cancel](#) [Create route table](#)

Public route table is created.

🔍 Search [Alt+S] United States (N. Virginia) 📍

VPC > Route tables > rtb-050c67bf656c12775 > Edit subnet associations

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (1/2)

Filter subnet associations

☒ Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☒ public-subnet	subnet-0353c20696b164aa9	10.0.0.0/24	-	Main (rtb-032ff6032490e77ec7)
☐ private-subnet	subnet-0ebd57b722fc44bf3	10.0.1.0/25	-	Main (rtb-032ff6032490e77ec7)

Selected subnets

subnet-0353c20696b164aa9 / public-subnet X

[Cancel](#) [Save associations](#)

Public association is added.

🔍 Search [Alt+S] United States (N. Virginia) 📍

VPC > Route tables > rtb-050c67bf656c12775 > Edit routes

Edit routes

Destination	Target	Status	Propagated
10.0.0.0/16	local	Active	No
Q 0.0.0.0/0 X	Internet Gateway	-	No

[Add route](#)

[Cancel](#) [Preview](#) [Save changes](#)

Internet gateway is added.

Create route table info

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

private-route

VPC
The VPC to use for this route table.

vpc-08ace5e1543fc306 (my-vpc)

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key **Value - optional**

Q Name X Q private-route X Remove

Add new tag

You can add 49 more tags.

Cancel Create route table

Private route table is created.

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (1/2)

Filter subnet associations

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
public-subnet	subnet-0353c20696b164aa9	10.0.0.0/24	-	rtb-050c67bf556c12775 / public-route
private-subnet	subnet-0ebd37b722fc44bf3	10.0.1.0/25	-	Main (rtb-02dfe032490c77ec7)

Selected subnets

subnet-0ebd37b722fc44bf3 / private-subnet X

Cancel Save associations

Edit routes

Destination	Target	Status	Propagated
10.0.0.0/16	local	Active	No
0.0.0.0/0	NAT Gateway	-	No

Remove

Add route

Cancel Preview Save changes

NAT gate way is added to private.

Updated routes for rtb-0f7d3757240a58364 / private-route successfully

Route tables (4) info

Find route tables by attribute or tag

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC	Owner ID
-	rtb-08bfee1b4572c73906	-	-	Yes	vpc-0c1e9f84aa49c9b9e	163083073206
-	rtb-02dfe032490c77ec7	-	-	Yes	vpc-08ace5e1543fc306 my-vpc	163083073206
public-route	rtb-050c67bf556c12775	subnet-0353c20696b164...	-	No	vpc-08ace5e1543fc306 my-vpc	163083073206
private-route	rtb-0f7d3757240a58364	subnet-0ebd37b722fc44...	-	No	vpc-08ace5e1543fc306 my-vpc	163083073206

Both public and private route table are added.

Launch an instance

It seems like you may be new to launching instances in EC2. Take a walkthrough to learn about EC2, how to launch instances and about best practices.

Do not show me this message again Take a walkthrough

Launch an instance

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

Name: public Add additional tags

Application and OS Images (Amazon Machine Image)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or browse for AMIs if you don't see what you are looking for below.

Search our full catalog including 1000s of application and OS images

Recents Quick Start

Amazon Linux macOS Ubuntu Windows Red Hat SUSE Linux Debian

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI
ami-09e6f87a7903347c (64-bit (x86), x86_64) / ami-0db3bcb0a66b988 (64-bit (arm), aarch64)
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Summary

Number of instances: 1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.7.2...read more
ami-09e6f87a7903347c

Virtual server type (instance type)
t2.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GiB of snapshots, and 100 GB of bandwidth to the internet.

Cancel Launch instance Preview code

Public instance is created.

Launch an instance

Network settings

VPC - required
vpc-0ba6ce5e1543fc306 (my-vpc) 10.0.0.0/16

Subnet
subnet-055c2d696b164aa9 public-subnet VPC: vpc-0ba6ce5e1543fc306 Owner: 163083073206 Availability Zone: us-east-1a Zone type: Availability Zone IP addresses available: 250 CIDR: 10.0.0.0/24 Create new subnet

Auto-assign public IP
Enable

Additional charges apply when outside of free tier allowance

Firewall (security groups)
Create security group Select existing security group

Security group name - required
public-new

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and - (hyphen) and ! (exclamation point).

Description - required
public-new/created 2025-06-17T09:59:49.039Z

Inbound Security Group Rules

Security group rule 1 (TCP, 22, 0.0.0.0/0)

Type: ssh Protocol: TCP Port range: 22

Source type: Anywhere Source: 0.0.0.0/0 Description - optional: e.g. SSH for admin desktop

Security group rule 2 (ICMP, All, 0.0.0.0/0)

Type: All ICMP - IPv4 Protocol: ICMP Port range: All

Source type: Anywhere Source: 0.0.0.0/0 Description - optional: e.g. SSH for admin desktop

Security group rule 3 (TCP, 80, 0.0.0.0/0)

Type: HTTP Protocol: TCP Port range: 80

Source type: Anywhere Source: 0.0.0.0/0 Description - optional: e.g. SSH for admin desktop

Summary

Number of instances: 1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.7.2...read more
ami-09e6f87a7903347c

Virtual server type (instance type)
t2.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GiB of snapshots, and 100 GB of bandwidth to the internet.

Cancel Launch instance Preview code

Necessary Network setting is updated.

Launch an instance

Inbound Security Group Rules

Security group rule 1 (TCP, 22, 0.0.0.0/0)

Type: ssh Protocol: TCP Port range: 22

Source type: Anywhere Source: 0.0.0.0/0 Description - optional: e.g. SSH for admin desktop

Security group rule 2 (ICMP, All, 0.0.0.0/0)

Type: All ICMP - IPv4 Protocol: ICMP Port range: All

Source type: Anywhere Source: 0.0.0.0/0 Description - optional: e.g. SSH for admin desktop

Security group rule 3 (TCP, 80, 0.0.0.0/0)

Type: HTTP Protocol: TCP Port range: 80

Source type: Anywhere Source: 0.0.0.0/0 Description - optional: e.g. SSH for admin desktop

Summary

Number of instances: 1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.7.2...read more
ami-09e6f87a7903347c

Virtual server type (instance type)
t2.micro

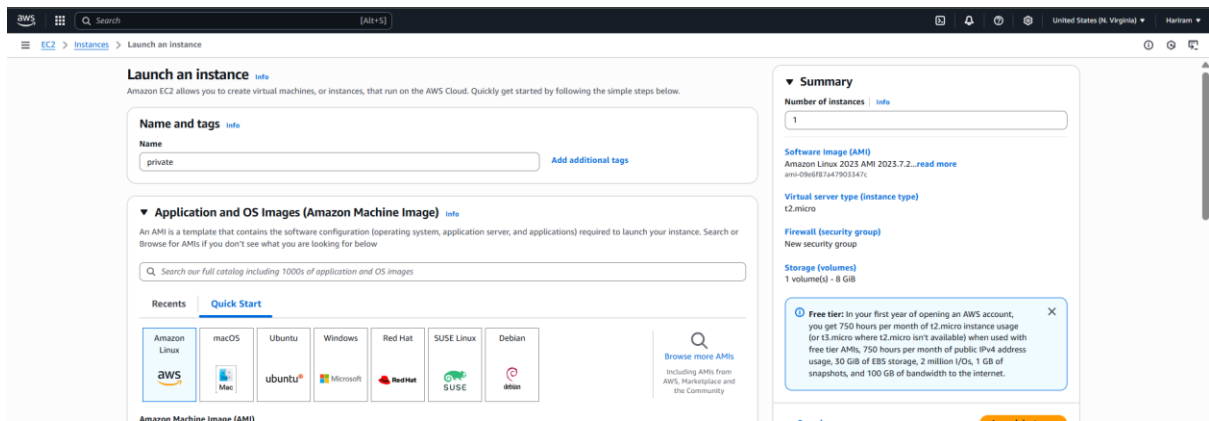
Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

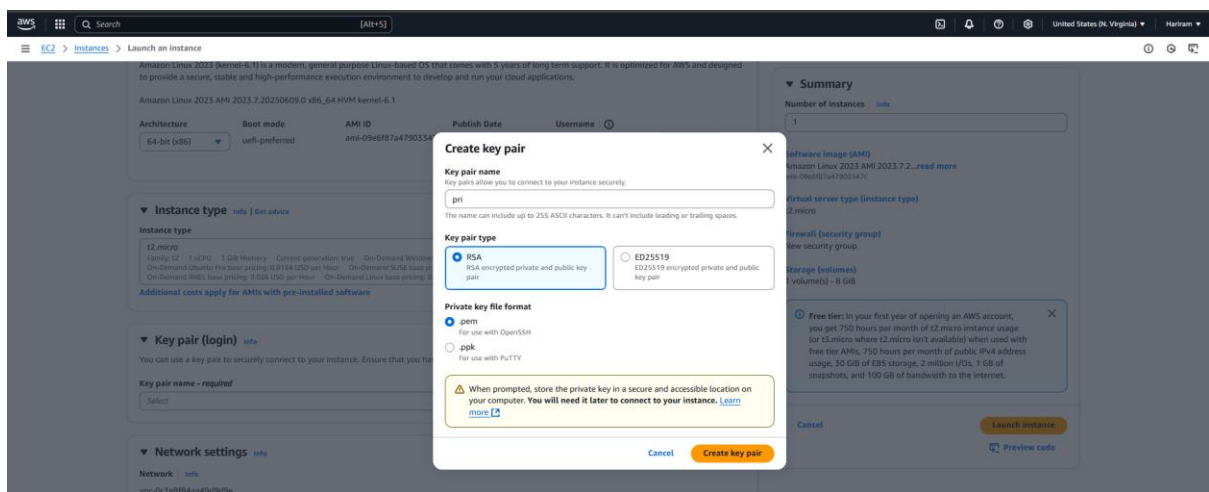
Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GiB of snapshots, and 100 GB of bandwidth to the internet.

Cancel Launch instance Preview code

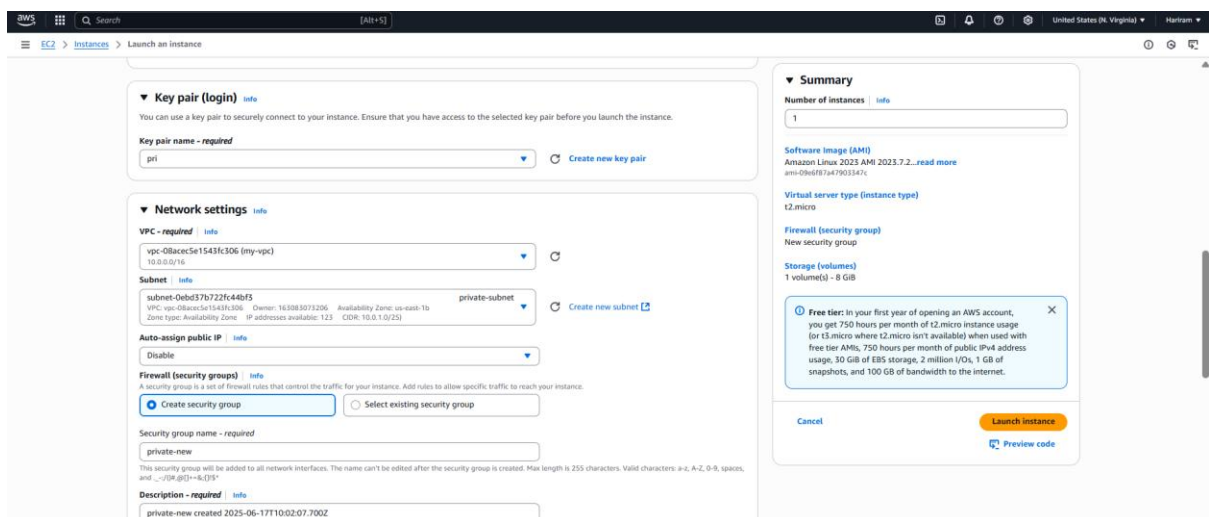
Necessary ports are added in security group.



Private instance is created.



New key is created for private.

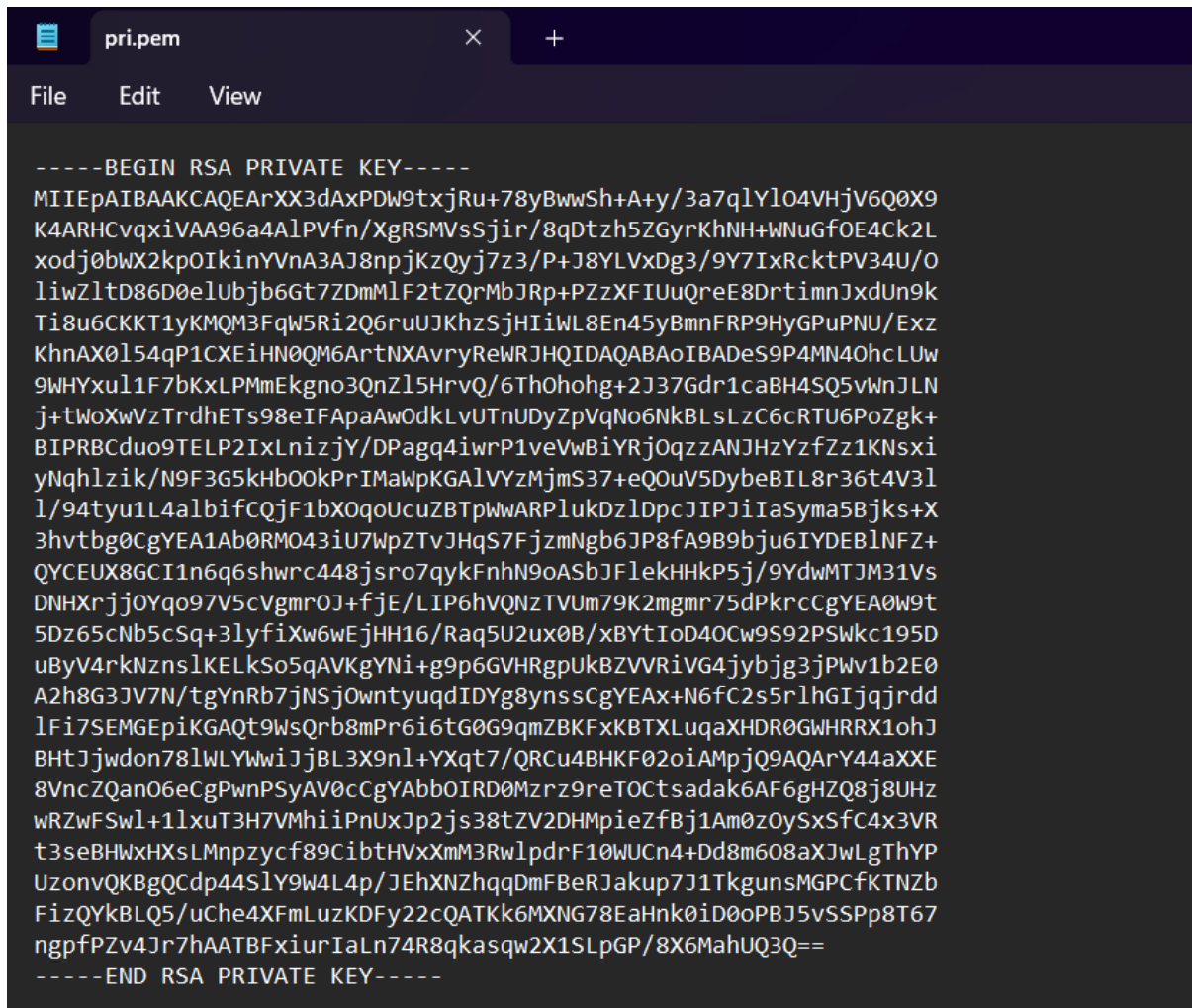


Security is created.

Web server is live.

```
[root@ip-10-0-0-167 ~]# systemctl restart httpd
[root@ip-10-0-0-167 ~]# systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[root@ip-10-0-0-167 ~]# vim pri.pem
[root@ip-10-0-0-167 ~]# |
```

Private key is add inside the public instance to give internet access to private instance through NAT gateway.

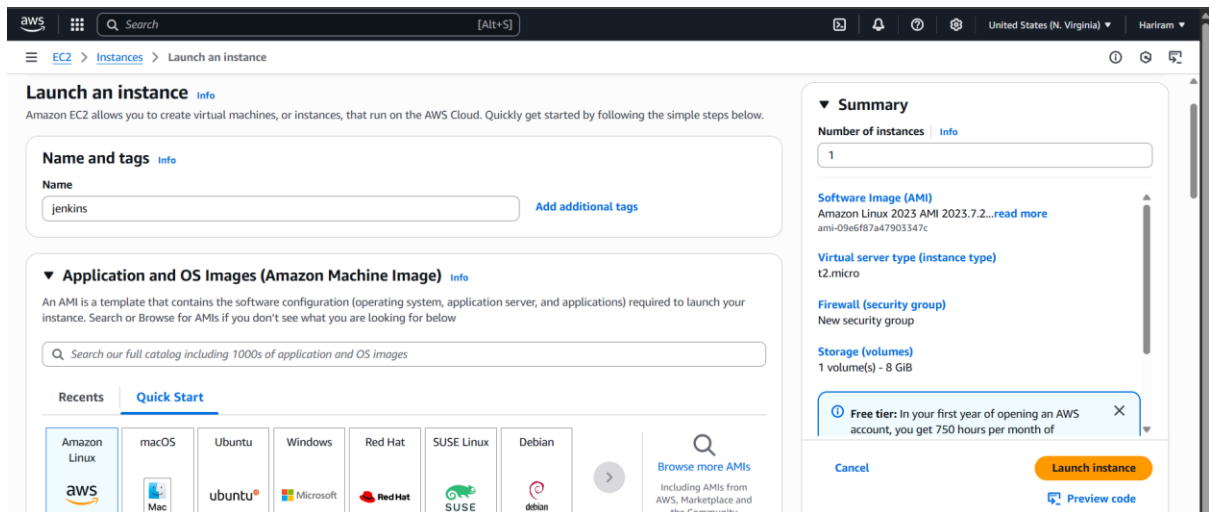


```
pri.pem
File Edit View

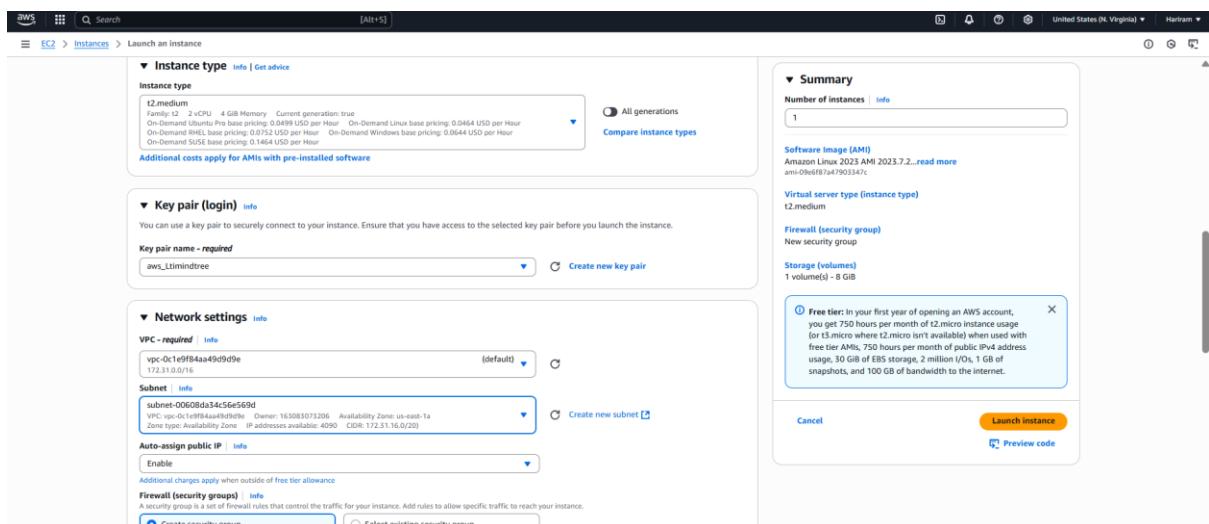
-----BEGIN RSA PRIVATE KEY-----
MIIEpAIBAAKCAQEARXX3dAxPDW9txjRu+78yBwwSh+A+y/3a7qlYl04VHjV6Q0X9
K4ARHCvqxiVAA96a4AlPVfn/XgRSMVsSjir/8qDtzh5ZGyrKhNH+WNuGf0E4Ck2L
xodj0bWx2kp0IkinYVnA3AJ8npjKzQyj7z3/P+J8YLvxDg3/9Y7IxRcktPV34U/O
liwZltd86D0e1Ubjb6Gt7ZDmMlF2tZQRmbJRp+PZzXFUuQreE8DrtimnJxdUn9k
Ti8u6CKKT1yKMQM3FqW5Ri2Q6ruUJKhzSjHIiWL8En45yBmnFRP9HyGPuPNU/Exz
KhnAX0l54qP1CXEiHN0QM6ArtNXAvryReWRJHQIDAQABaoIBADeS9P4MN40hcLUw
9WHYxul1F7bKxLPMmEkgn03QnZl5HrvQ/6Th0hohg+2J37Gdr1caBH4SQ5vWnJLN
j+tWoXwVzTrdHETs98eIFApaAwOdkLVUTnUDyZpVqNo6NkBLsLzC6cRTU6PoZgk+
BIPRBCduo9TELP2IXLnizjY/DPagq4iwrP1veVwBiYRj0qzzANJHzYzfZz1KNSxi
yNqhlzik/N9F3G5kHb00kPrImawpKGAlVYzMjms37+eQ0uV5DybeBIL8r36t4V3l
l/94tyu1L4albifCQjF1bXOqoUcuZBTpWwARPlukDz1DpcJIPjiIaSyma5Bjks+X
3hvtbg0CgYEA1Ab0RMO43iU7WpZTvjHqS7FjzmNgb6JP8fA9B9bj6IYDEB1NFZ+
QYCEUX8GCI1n6q6shwrc448jsro7qykFnhN9oASbJflekHHKp5j/9YdwMTJM31Vs
DNHxrjj0Yqo97V5cVgmr0J+fjE/LIP6hVQNzTVUm79K2mgmr75dPkrcCgYEA0W9t
5Dz65cNb5cSq+3lyfiXw6wEjHH16/Raq5U2ux0B/xBYtIoD40Cw9S92PSWkc195D
uByV4rkNznslKELkSo5qAVKgYni+g9p6GVHRgpUkBVVRiVG4jybjg3jPWv1b2E0
A2h8G3JV7N/tgYnRb7jNSjOwntyudIDYg8ynssCgYEAx+N6fC2s5r1hGIjqrdd
lFi7SEMGEpiKGAQt9WsQrb8mPr6i6tG0G9qmZBKfXKBTXLuqaXHDR0GWHRRX1ohJ
BHtJjwdon78lWLYWwiJjBL3X9nl+YXqt7/QRcu4BHKF02oiAmpjQ9AAQArY44aXXE
8VncZQan06eCgPwnPSyAV0cCgYAbboIRD0Mzrz9reTOctsadak6AF6gHZQ8j8UHz
wRZwFswl+1lxuT3H7VMhiiPnUxJp2js38tZV2DHmpieZfBj1Am0zOySxSfC4x3VR
t3seBHWxHXsLMnpzycf89CibthVxXmM3RwlpdrF10WUCn4+Dd8m608aXJwLgThYP
UzonvQKBgQCdp44Sly9W4L4p/JEhXNZhqDmFBeRJakup7J1TkgunsMGPCfKTNZb
FizQYkBLQ5/uChe4XFmLuzKDFy22cQATKk6MXNG78EaHnk0iD0oPBj5vSSPp8T67
ngpfPZv4Jr7haATBFxiurIaLn74R8qkasqw2X1SLpGP/8X6MahUQ3Q==
-----END RSA PRIVATE KEY-----
```

This is the private key.

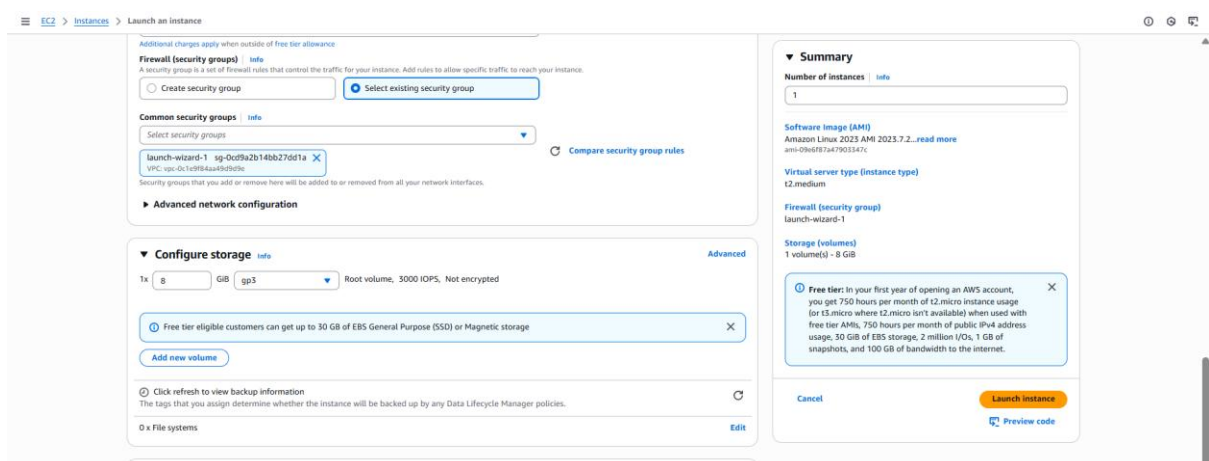
4) Install and configure Jenkins on Linux server and create a webhook with GitHub repository. Then build a Java base application using build tool maven. Save out put into pipeline



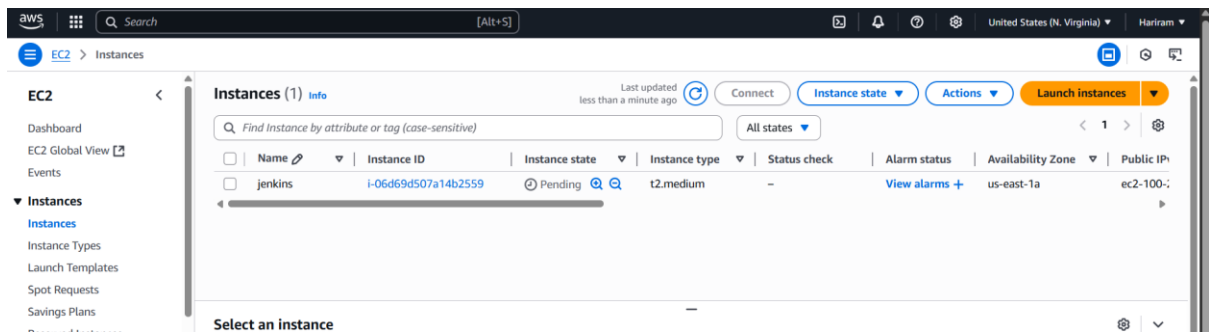
Jenkins instance is created.



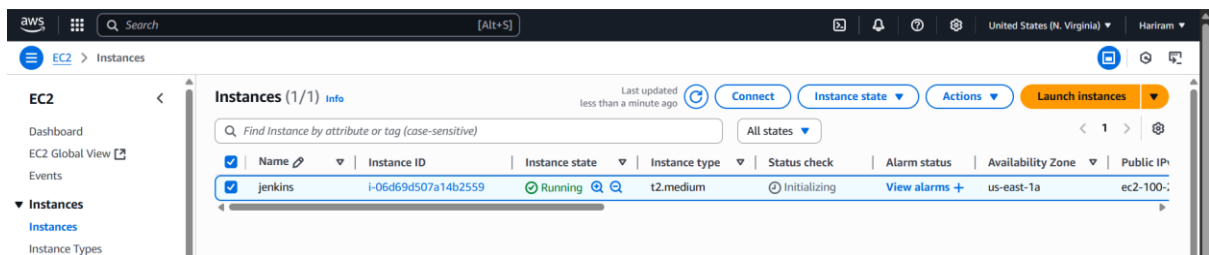
T2.medium is selected .



Existing security group is selected.



Jenkins EC2 created.



Jenkins Instance is running successfully.

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\10835505\OneDrive - LTIMindtree\Desktop> ssh -i "aws_Ltimindtree.pem" ec2-user@ec2-100-27-196-87.compute-1.amazonaws.com
The authenticity of host 'ec2-100-27-196-87.compute-1.amazonaws.com [100.27.196.87]' can't be established.
ED25519 key fingerprint is SHA256:mF864eRaTP92N8pBuHWpmlZBH7Ztx/H5jUe79xbYdPE.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-100-27-196-87.compute-1.amazonaws.com' (ED25519) to the list of known hosts.
```



```
#_
#_#####_   Amazon Linux 2023
#_#####_
#_####|
#_#/
#_#V#I-->      https://aws.amazon.com/linux/amazon-linux-2023
#_#
#_#
#_#m/'
```

```
[ec2-user@ip-172-31-19-94 ~]$ sudo su -
[root@ip-172-31-19-94 ~]# hostnamectl set-hostname jenkins.example.com
[root@ip-172-31-19-94 ~]# bash
[root@jenkins ~]# yum update -y
Amazon Linux 2023 Kernel Livepatch repository                      180 kB/s | 17 kB          00:00
No match for argument: -y
Error: No packages marked for upgrade.
[root@jenkins ~]#
```

Yum update is done.

```
[root@jenkins ~]# wget -O /etc/yum.repos.d/jenkins.repo \
https://pkg.jenkins.io/redhat-stable/jenkins.repo
--2025-06-17 09:21:17-- https://pkg.jenkins.io/redhat-stable/jenkins.repo
Resolving pkg.jenkins.io (pkg.jenkins.io)... 146.75.34.133, 2a04:4e42:78::645
Connecting to pkg.jenkins.io (pkg.jenkins.io)|146.75.34.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 85
Saving to: '/etc/yum.repos.d/jenkins.repo'

/etc/yum.repos.d/jenkins.repo 100%[=====>] 85 --.-KB/s in 0s

2025-06-17 09:21:18 (5.62 MB/s) - '/etc/yum.repos.d/jenkins.repo' saved [85/85]
```

Necessary things to run Jenkins are installed.

```
[root@jenkins ~]# rpm --import https://pkg.jenkins.io/redhat-stable/jenkins.io-2023.key
[root@jenkins ~]# yum upgrade
Jenkins-stable                               714 kB/s | 31 kB      00:00
Dependencies resolved.
Nothing to do.
Complete!
[root@jenkins ~]# yum install java-17-amazon-corretto -y
Last metadata expiration check: 0:00:12 ago on Tue Jun 17 09:21:57 2025.
Dependencies resolved.
=====
Package                                Architecture      Version                                Repository          Size
=====
Installing:
  java-17-amazon-corretto                x86_64            1:17.0.15+6-1.amzn2023.1             amazonlinux          213 k
Installing dependencies:
  alsa-lib                                x86_64            1.2.7.2-1.amzn2023.0.2               amazonlinux          504 k
=====
```

```
[root@jenkins ~]# yum install jenkins -y
Last metadata expiration check: 0:00:51 ago on Tue Jun 17 09:21:57 2025.
Dependencies resolved.
=====
Package                                Architecture      Version                                Repository          Size
=====
Installing:
  jenkins                                noarch            2.504.2-1.1                           jenkins              90 M
=====
Transaction Summary
=====
Install 1 Package

Total download size: 90 M
Installed size: 90 M
Downloading Packages:
jenkins-2.504.2-1.1.noarch.rpm           10 MB/s | 90 MB      00:08
-----
Total                                     10 MB/s | 90 MB      00:08
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing                               :                               1/1
  Running scriptlet: jenkins-2.504.2-1.1.noarch : 1/1
  Installing        : jenkins-2.504.2-1.1.noarch : 1/1
  Running scriptlet: jenkins-2.504.2-1.1.noarch : 1/1
  Verifying         : jenkins-2.504.2-1.1.noarch : 1/1
Installed:
  jenkins-2.504.2-1.1.noarch
Complete!
```

Jenkins is installed

```
[root@jenkins ~]# yum install git -y
Last metadata expiration check: 0:05:53 ago on Tue Jun 17 09:21:57 2025.
Dependencies resolved.
=====
Package                                Architecture      Version                                Repository          Size
=====
Installing:
  git                                    x86_64            2.47.1-1.amzn2023.0.3                 amazonlinux          52 k
Installing dependencies:
  git-core                               x86_64            2.47.1-1.amzn2023.0.3                 amazonlinux          4.5 M
  git-core-doc                           noarch            2.47.1-1.amzn2023.0.3                 amazonlinux          2.8 M
  perl-Error                             noarch            1:0.17029-5.amzn2023.0.2              amazonlinux          41 k
  perl-File-Find                         noarch            1.37-477.amzn2023.0.7                 amazonlinux          25 k
  perl-Git                               noarch            2.47.1-1.amzn2023.0.3                 amazonlinux          40 k
  perl-TermReadKey                       x86_64            2.38-9.amzn2023.0.2                   amazonlinux          36 k
  perl-lib                               x86_64            0.65-477.amzn2023.0.7                 amazonlinux          15 k
=====
Transaction Summary
=====
Install 8 Packages

Total download size: 7.5 M
Installed size: 37 M
Downloading Packages:
(1/8): git-2.47.1-1.amzn2023.0.3.x86_64.rpm           1.4 MB/s | 52 kB      00:00
(2/8): git-core-doc-2.47.1-1.amzn2023.0.3.noarch.rpm   39 MB/s | 2.8 MB      00:00
-----
```

Git is installed.

```
[root@jenkins ~]# yum install maven
Last metadata expiration check: 0:06:19 ago on Tue Jun 17 09:21:57 2025.
Dependencies resolved.
=====
Package                               Architecture Version                               Repository                               Size
=====
Installing:
maven                                noarch    1:3.8.4-3.amzn2023.0.5                amazonlinux                               18 k
Installing dependencies:
apache-commons-cli                   noarch    1.5.0-3.amzn2023.0.3                  amazonlinux                               76 k
apache-commons-codec                 noarch    1.15-6.amzn2023.0.3                   amazonlinux                               303 k
apache-commons-io                    noarch    1:2.8.0-7.amzn2023.0.5                 amazonlinux                               283 k
apache-commons-lang3                 noarch    3.12.0-7.amzn2023.0.3                 amazonlinux                               559 k
atinject                             noarch    1.0.5-3.amzn2023.0.3                  amazonlinux                               23 k
cdi-api                              noarch    2.0.2-6.amzn2023.0.3                  amazonlinux                               54 k
google-guice                         noarch    4.2.3-8.amzn2023.0.6                  amazonlinux                               473 k
guava                                noarch    31.0.1-3.amzn2023.0.6                  amazonlinux                               2.4 M
httpcomponents-client                noarch    4.5.13-4.amzn2023.0.4                 amazonlinux                               657 k
httpcomponents-core                  noarch    4.4.13-6.amzn2023.0.3                 amazonlinux                               632 k
jakarta-annotations                 noarch    1.3.5-13.amzn2023.0.3                 amazonlinux                               46 k
jansi                                x86_64    2.4.0-3.amzn2023.0.3                  amazonlinux                               113 k
java-17-amazon-corretto-devel        x86_64    1:17.0.15+6-1.amzn2023.1              amazonlinux                               142 k
jcl-over-slf4j                       noarch    1.7.32-3.amzn2023.0.4                 amazonlinux                               25 k
jsoup                                noarch    1.16.1-4.amzn2023.0.2                 amazonlinux                               433 k
jsr-305                              noarch    3.0.2-5.amzn2023.0.4                  amazonlinux                               32 k
maven-amazon-corretto17              noarch    1:3.8.4-3.amzn2023.0.5                 amazonlinux                               9.4 k
maven-lib                            noarch    1:3.8.4-3.amzn2023.0.5                 amazonlinux                               1.5 M
maven-resolver                       noarch    1.1.7.3-3.amzn2023.0.4                amazonlinux                               557 k
maven-shared-utils                   noarch    3.3.4-4.amzn2023.0.3                  amazonlinux                               152 k
=====
```

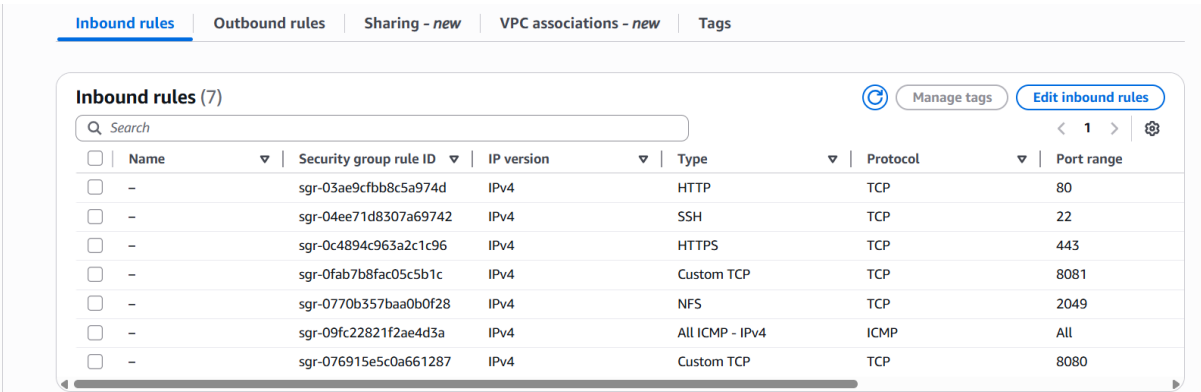
Maven is installed.

```
[root@jenkins ~]# systemctl enable jenkins
Created symlink /etc/systemd/system/multi-user.target.wants/jenkins.service → /usr/lib/systemd/system/jenkins.service.
[root@jenkins ~]# systemctl start jenkins
[root@jenkins ~]# systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: disabled)
   Active: active (running) since Tue 2025-06-17 09:31:30 UTC; 5s ago
     Main PID: 4133 (java)
       Tasks: 53 (limit: 4656)
      Memory: 886.8M
         CPU: 19.072s
    CGroup: /system.slice/jenkins.service
            └─4133 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --http=

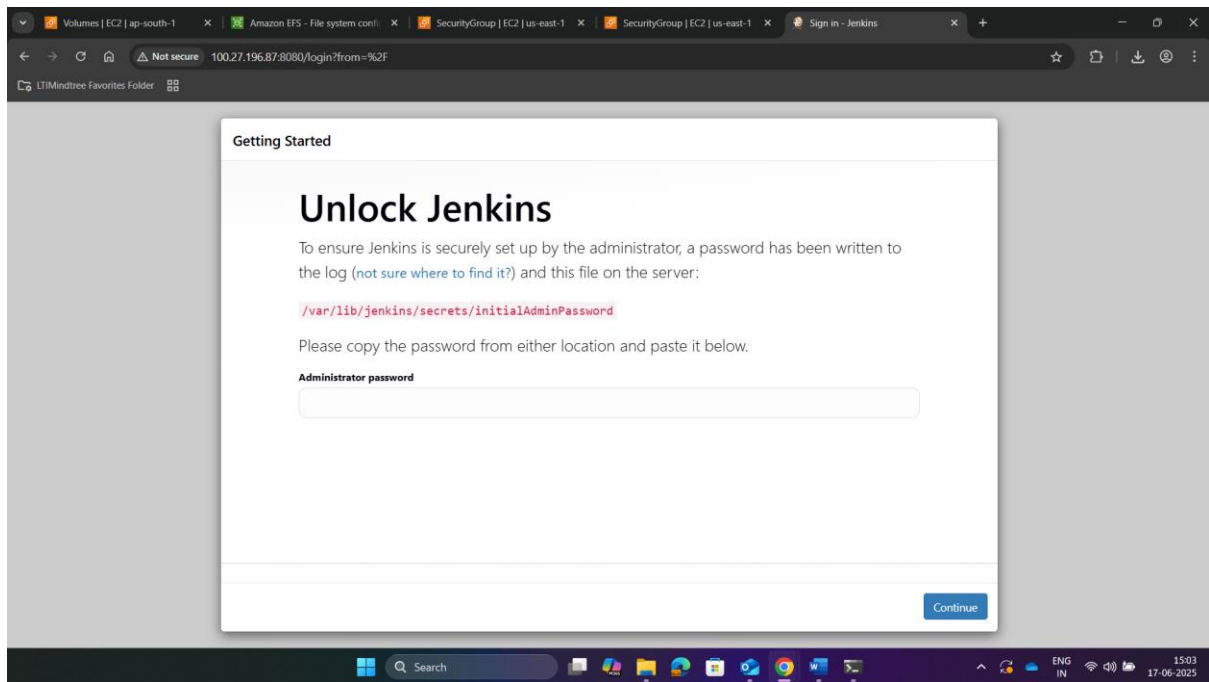
Jun 17 09:31:26 jenkins.example.com jenkins[4133]: 675f0bf431ff461cab9b3cb28f879572
Jun 17 09:31:26 jenkins.example.com jenkins[4133]: This may also be found at: /var/lib/jenkins/secrets/initialAdminPassword
Jun 17 09:31:26 jenkins.example.com jenkins[4133]: *****
Jun 17 09:31:26 jenkins.example.com jenkins[4133]: *****
Jun 17 09:31:26 jenkins.example.com jenkins[4133]: *****
Jun 17 09:31:30 jenkins.example.com jenkins[4133]: 2025-06-17 09:31:30.509+0000 [id=33] INFO jenkins.InitReactorRunn
Jun 17 09:31:30 jenkins.example.com jenkins[4133]: 2025-06-17 09:31:30.522+0000 [id=23] INFO hudson.lifecycle.Lifecyc
Jun 17 09:31:30 jenkins.example.com systemd[1]: Started jenkins.service - Jenkins Continuous Integration Server.
Jun 17 09:31:30 jenkins.example.com jenkins[4133]: 2025-06-17 09:31:30.712+0000 [id=49] INFO h.m.DownloadService$Dow
Jun 17 09:31:30 jenkins.example.com jenkins[4133]: 2025-06-17 09:31:30.713+0000 [id=49] INFO hudson.util.Retrier$sta

[root@jenkins ~]# |
```

Jenkins is started.



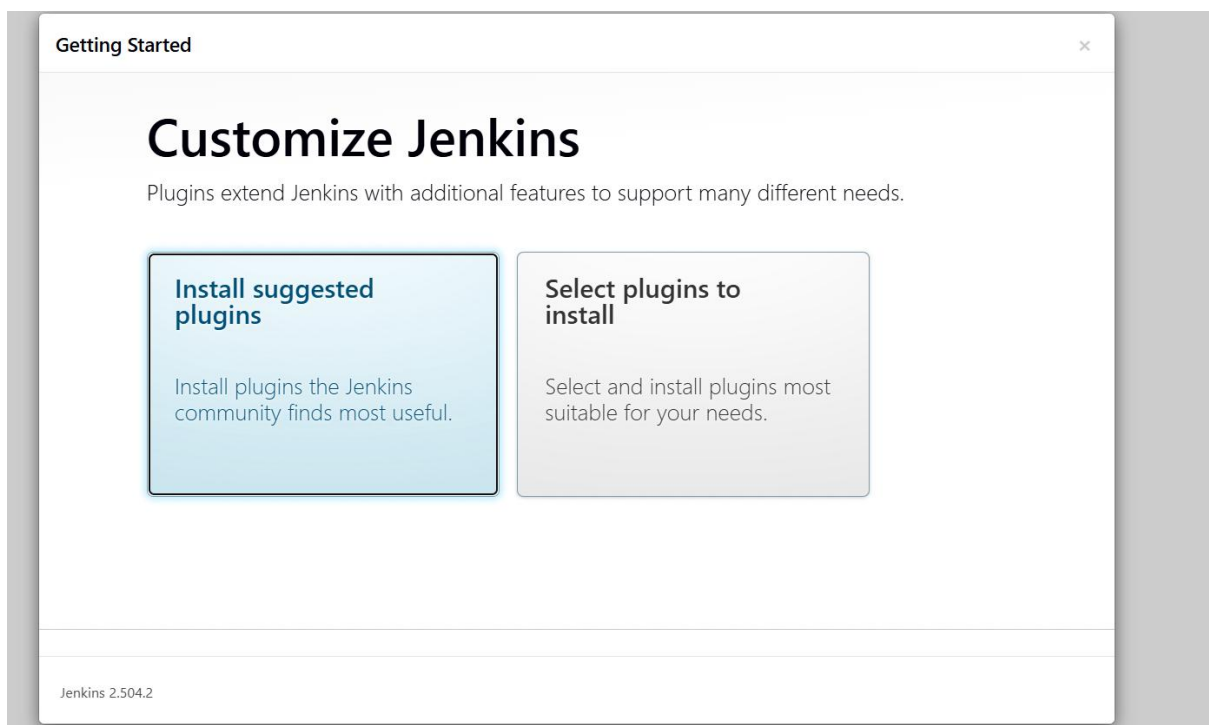
8080 port is added to Inbound.



Jenkins is accessible.

```
[root@jenkins ~]# cat /var/lib/jenkins/secrets/initialAdminPassword
675f0bf431ff461cab9b3cb28f879572
[root@jenkins ~]#
```

Key is generated.



Getting Started

Create First Admin User

Username

admin

Password

....

Confirm password

....

Full name

Hariram

E-mail address

hariramdevans@outlook.com

Jenkins 2.504.2

[Skip and continue as admin](#)

[Save and Continue](#)

Jenkins's admin is done.

Hairam24 / java_project

Type [Z] to search

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java_project Public

Pin Watch 0 Fork 0 Star 0

main 1 Branch 0 Tags

Go to file t Add file <> Code

Hairam24 Update index.jsp 239bbe5 · last week 11 Commits

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webapp Update index.jsp last week

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pom.xml edit 11 months ago

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Packages

Java project in git hub.

Jenkins

log out

Dashboard Manage Jenkins Plugins

Plugins

Updates

Available plugins

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Download progress

maven

Install

Released

☒

Maven Integration 3.26

Build Tools

This plugin provides a deep integration between Jenkins and Maven. It adds support for automatic triggers between projects depending on SNAPSHOTS as well as the automated configuration of various Jenkins publishers such as Junit.

1 mo 22 days ago

☐

Config File Provider 908.v0461fcc2b_9d1

Groovy-related External Site/Tool Integrations Maven

Ability to provide configuration files (e.g. settings.xml for maven, XML, groovy, custom files,...) loaded through the UI which will be copied to the job workspace.

1 mo 19 days ago

☐

Jira 3.17

External Site/Tool Integrations Maven Jira

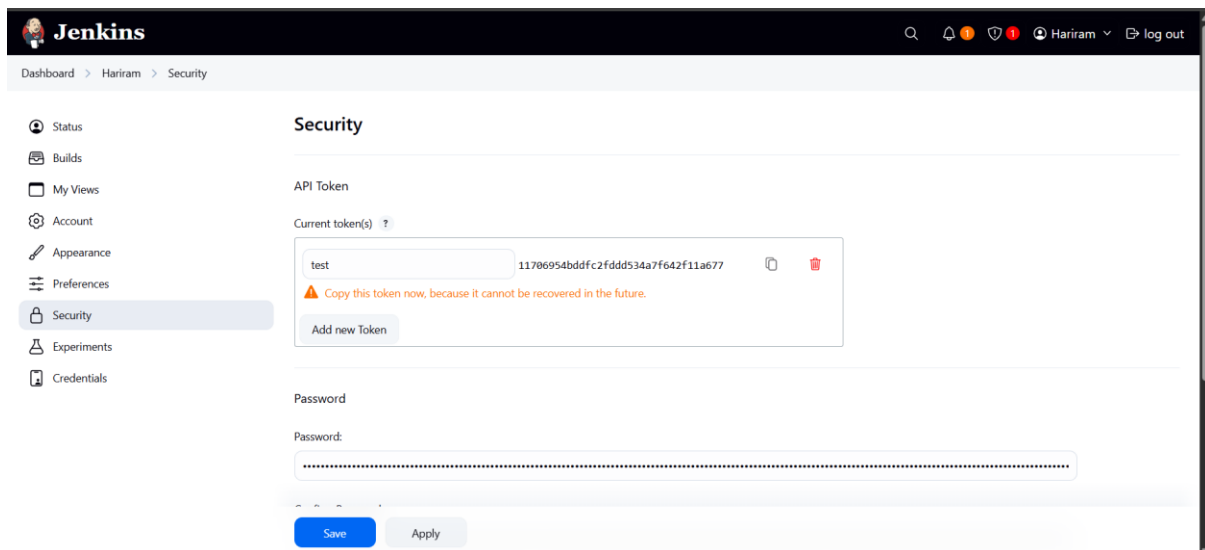
This plugin integrates Jenkins to Atlassian Jira.

11 days ago

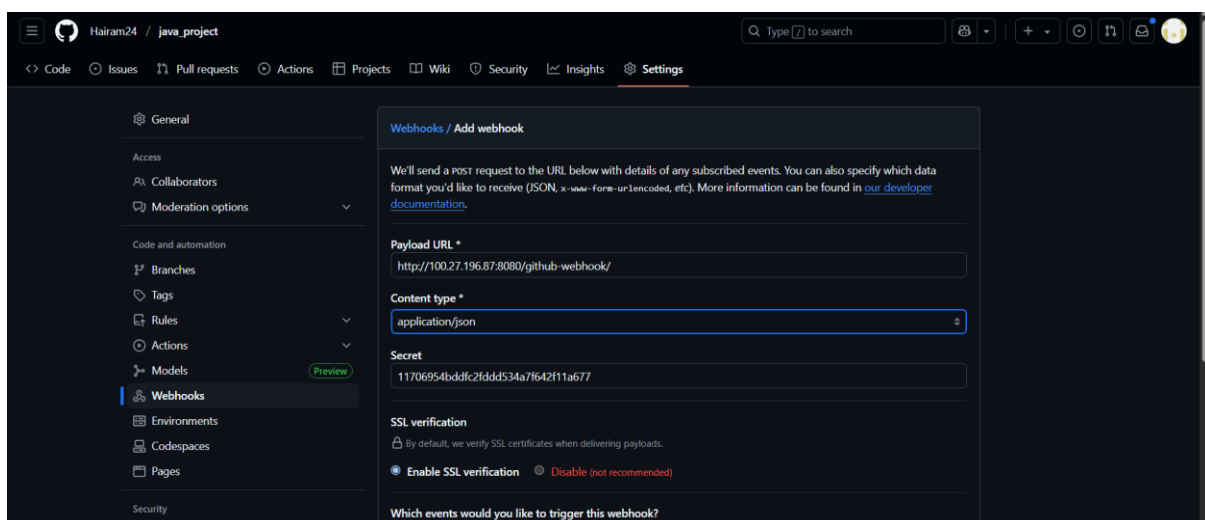
☐

Pipeline Maven Integration 1538.v40eb_89e33cc6

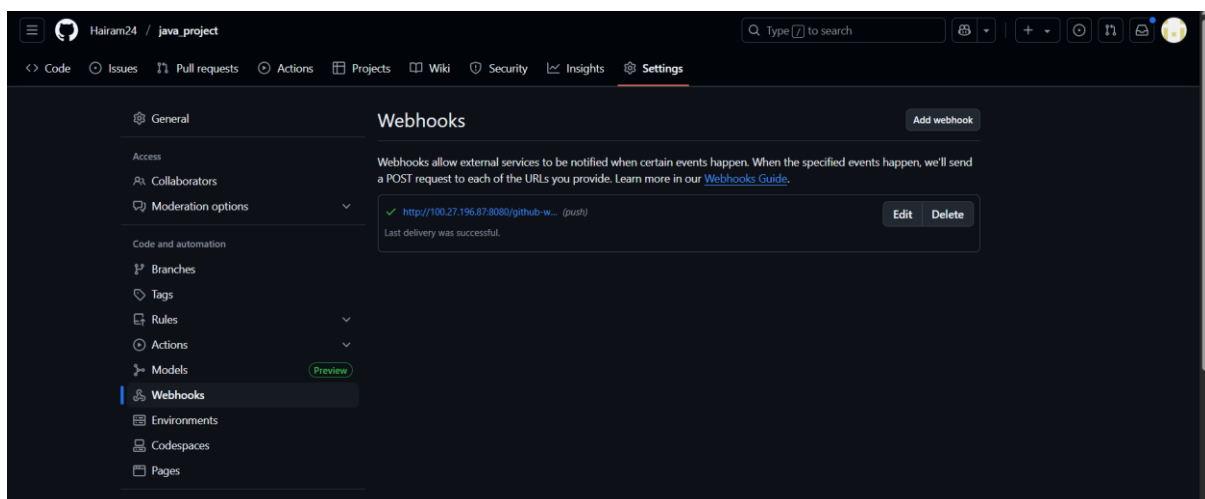
Maven plugin is installed.



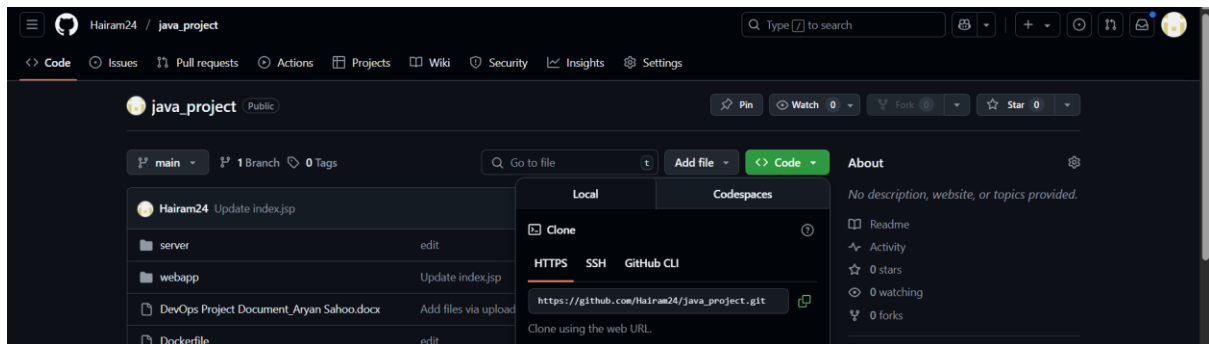
Token is generated for webhook.



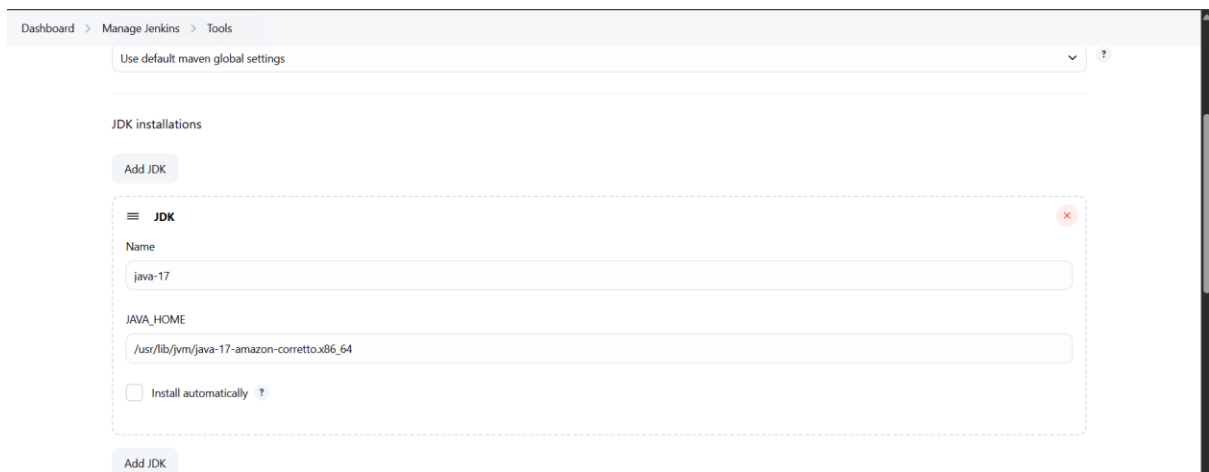
Webhook is created .



Webhook set up is done.



```
[root@jenkins ~]# mvn -v
Apache Maven 3.8.4 (Red Hat 3.8.4-3.amzn2023.0.5)
Maven home: /usr/share/maven
Java version: 17.0.15, vendor: Amazon.com Inc., runtime: /usr/lib/jvm/java-17-amazon-corretto.x86_64
Default locale: en, platform encoding: UTF-8
OS name: "linux", version: "6.1.140-154.222.amzn2023.x86_64", arch: "amd64", family: "unix"
[root@jenkins ~]#
```



Java path is added.

Dashboard > Manage Jenkins > Tools

Maven installations

Add Maven

Maven

Name

maven

MAVEN_HOME

/usr/share/maven

☐ Install automatically ?

Add Maven

Save Apply

Maven path is added.

Jenkins

Dashboard > All > New Item

New Item

Enter an item name

java_maven

Select an item type

- Freestyle project**
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
- Maven project**
Build a maven project. Jenkins takes advantage of your POM files and drastically reduces the configuration.
- Pipeline**
Orchestrates long-running activities that can span multiple build agents. Suitable for building pipelines (formerly known as workflows) and/or organizing complex activities that do not easily fit in free-style job type.
- Multi-configuration project**
Suitable for projects that need a large number of different configurations, such as testing on multiple environments.

OK

New item is created as maven project.

Dashboard > java_maven > Configuration

Configure

- General
- Source Code Management**
- Triggers
- Environment
- Pre Steps
- Build
- Post Steps
- Build Settings
- Post-build Actions

Git ?

Repositories ?

Repository URL ?

https://github.com/Hairam24/java_project.git

Credentials ?

- none -

+ Add

Advanced

Add Repository

Branches to build ?

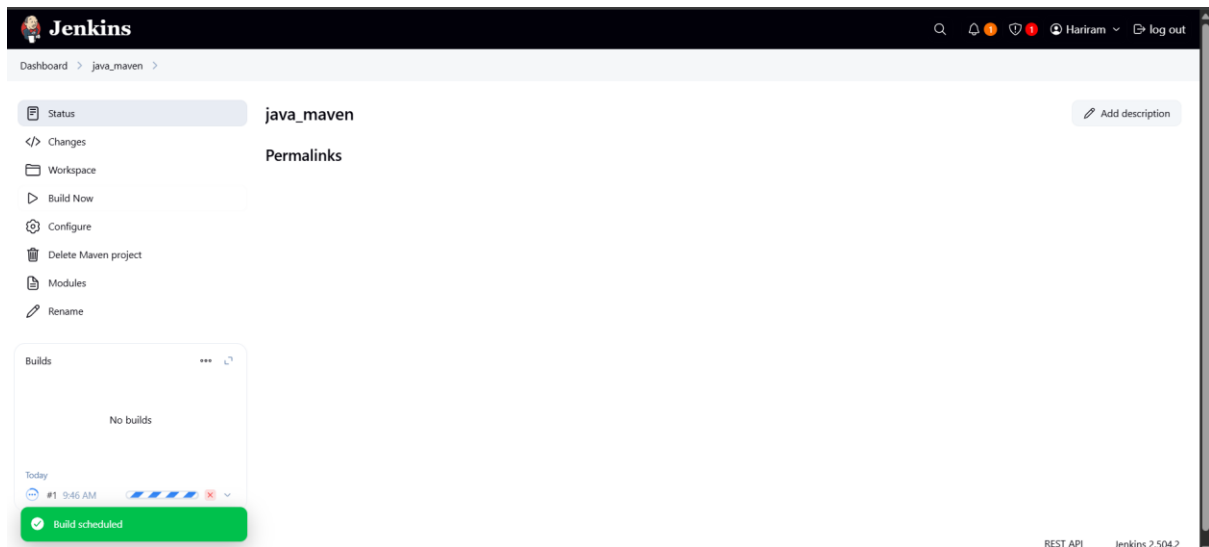
Branch Specifier (blank for 'any') ?

*/main

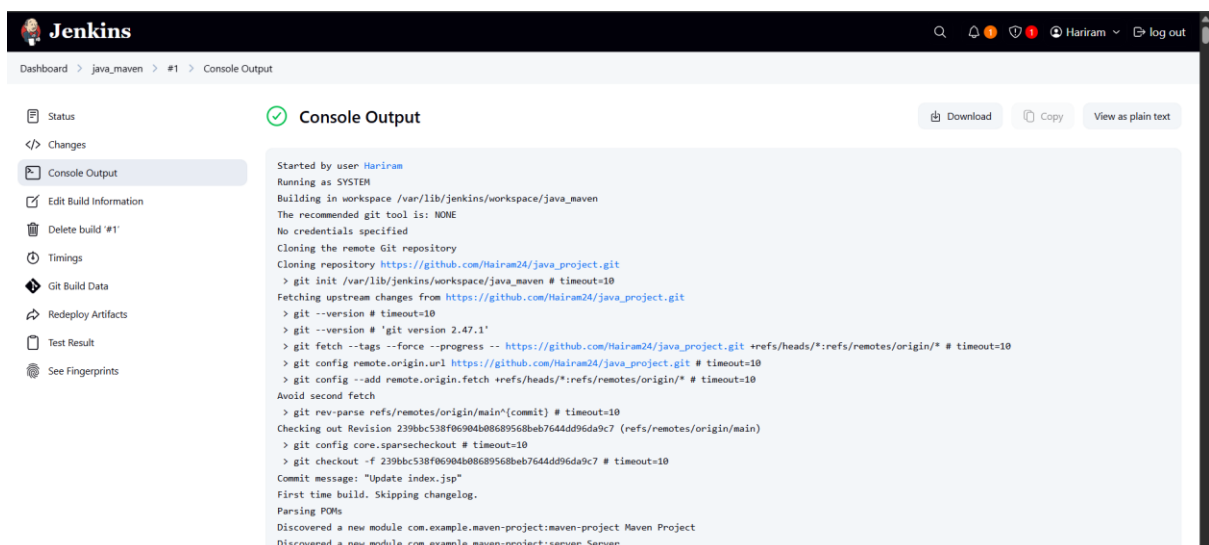
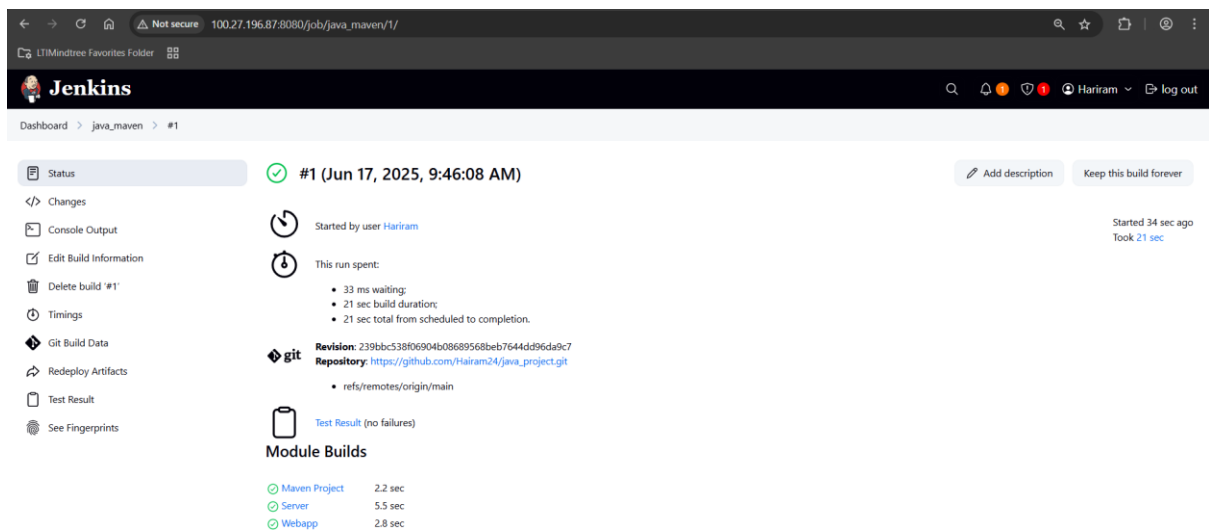
Add Branch

Save Apply

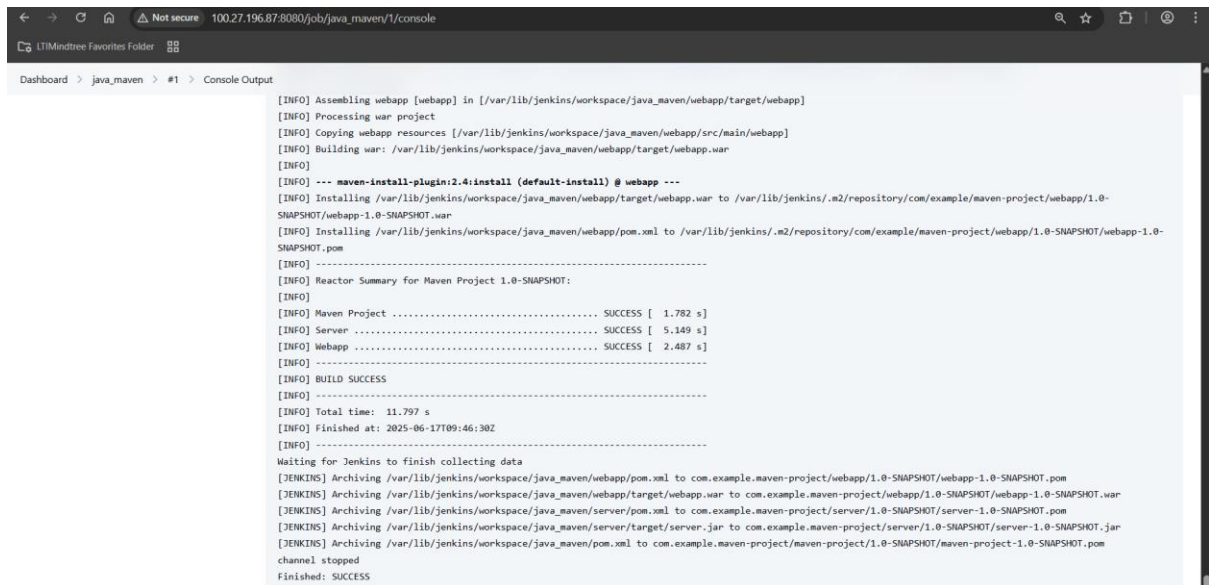
Git is connected.



Build successfully.



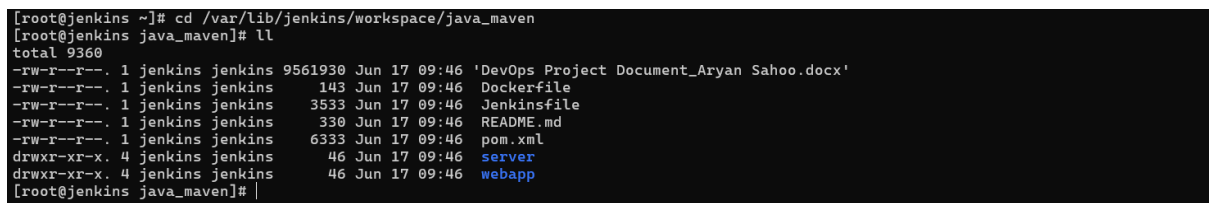
Var file is created successfully.



```
Dashboard > java_maven > #1 > Console Output

[INFO] Assembling webapp [webapp] in [/var/lib/jenkins/workspace/java_maven/webapp/target/webapp]
[INFO] Processing war project
[INFO] Copying webapp resources [/var/lib/jenkins/workspace/java_maven/webapp/src/main/webapp]
[INFO] Building war: /var/lib/jenkins/workspace/java_maven/webapp/target/webapp.war
[INFO]
[INFO] --- maven-install-plugin:2.4:install (default-install) @ webapp ---
[INFO] Installing /var/lib/jenkins/workspace/java_maven/webapp/target/webapp.war to /var/lib/jenkins/.m2/repository/com/example/maven-project/webapp/1.0-SNAPSHOT/webapp-1.0-SNAPSHOT.war
[INFO] Installing /var/lib/jenkins/workspace/java_maven/webapp/pom.xml to /var/lib/jenkins/.m2/repository/com/example/maven-project/webapp/1.0-SNAPSHOT/webapp-1.0-SNAPSHOT.pom
[INFO] -----
[INFO] Reactor Summary for Maven Project 1.0-SNAPSHOT:
[INFO]
[INFO] Maven Project ..... SUCCESS [ 1.782 s]
[INFO] Server ..... SUCCESS [ 5.149 s]
[INFO] Webapp ..... SUCCESS [ 2.487 s]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 11.797 s
[INFO] Finished at: 2025-06-17T09:46:30Z
[INFO] -----
Waiting for Jenkins to finish collecting data
[DEKINS] Archiving /var/lib/jenkins/workspace/java_maven/webapp/pom.xml to com.example.maven-project/webapp/1.0-SNAPSHOT/webapp-1.0-SNAPSHOT.pom
[DEKINS] Archiving /var/lib/jenkins/workspace/java_maven/webapp/target/webapp.war to com.example.maven-project/webapp/1.0-SNAPSHOT/webapp-1.0-SNAPSHOT.war
[DEKINS] Archiving /var/lib/jenkins/workspace/java_maven/server/pom.xml to com.example.maven-project/server/1.0-SNAPSHOT/server-1.0-SNAPSHOT.pom
[DEKINS] Archiving /var/lib/jenkins/workspace/java_maven/server/target/server.jar to com.example.maven-project/server/1.0-SNAPSHOT/server-1.0-SNAPSHOT.jar
[DEKINS] Archiving /var/lib/jenkins/workspace/java_maven/pom.xml to com.example.maven-project/maven-project/1.0-SNAPSHOT/maven-project-1.0-SNAPSHOT.pom
channel stopped
Finished: SUCCESS
```

Build run successful.



```
[root@jenkins ~]# cd /var/lib/jenkins/workspace/java_maven
[root@jenkins java_maven]# ll
total 9360
-rw-r--r--. 1 jenkins jenkins 9561930 Jun 17 09:46 'DevOps Project Document_Aryan Sahoo.docx'
-rw-r--r--. 1 jenkins jenkins 143 Jun 17 09:46 Dockerfile
-rw-r--r--. 1 jenkins jenkins 3533 Jun 17 09:46 Jenkinsfile
-rw-r--r--. 1 jenkins jenkins 330 Jun 17 09:46 README.md
-rw-r--r--. 1 jenkins jenkins 6333 Jun 17 09:46 pom.xml
drwxr-xr-x. 4 jenkins jenkins 46 Jun 17 09:46 server
drwxr-xr-x. 4 jenkins jenkins 46 Jun 17 09:46 webapp
[root@jenkins java_maven]#
```

output is saved into pipeline.